AI Memory Tool Comparison

This notebook benchmarks four memory layers for LLM applications — Zep, mem0, Supermemory, and Cognee — across setup, ingestion, and retrieval. The comparison is scoped to free-tier, beginner-friendly configurations, so results reflect what a developer gets without hitting a paywall, not each provider's full feature set.

Input data and retrieval tasks are held constant across all four providers to isolate differences in memory architecture rather than dataset artifacts. Each provider requires its own API key, set in a local `.env` file before running the notebook.

Before you run this notebook

- [ ] **API keys**: create a `.env` file in this notebook's directory with `ZEP_API_KEY`, `MEM0_API_KEY`, `SUPERMEMORY_API_KEY`, `COGNEE_API_KEY`, `COGNEE_URL`, and `OPENAI_API_KEY`.
- [ ] **Data path**: in section 2a ("Loading shared dataset"), update `DATA_ROOT` to point at your local copy of the `deterministic/` and `non_deterministic/` folders.

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1. Download & Setup

Each provider's setup ships with a basic ingestion/retrieval smoke test; those are omitted here to keep this section focused on installation and configuration.

Zep Installation

In [ ]:

```
%pip install zep-cloud
%pip install python-dotenv
import os
import uuid
from dotenv import load_dotenv
from zep_cloud.client import Zep
from zep_cloud.types import Message

load_dotenv()
zep_client = Zep(api_key=os.getenv("ZEP_API_KEY"))
```

mem0 Installation

In [ ]:

```
%pip install mem0ai
%pip install python-dotenv
%pip install mem0ai[nlp]
%pip install "mem0ai[extras]"
import os
from dotenv import load_dotenv
from mem0 import MemoryClient

load_dotenv()
mem0_client = MemoryClient(api_key=os.getenv("MEM0_API_KEY"))
```

Supermemory Installation

In [ ]:

```
%pip install supermemory
import os
from supermemory import Supermemory
from dotenv import load_dotenv


load_dotenv()
sm_client = Supermemory(api_key=os.getenv("SUPERMEMORY_API_KEY"))
```

Cognee Installation

In [ ]:

```
%pip install cognee
import os
from dotenv import load_dotenv
import cognee

load_dotenv()

async def main():
    await cognee.serve(
        url=os.getenv("COGNEE_URL"),
        api_key=os.getenv("COGNEE_API_KEY")
    )

await main()
```

2. Data Ingestion

The shared dataset is loaded once, then ingested into each provider individually. After each ingestion, we run a quick check to confirm the data landed correctly before moving on to Section 3: Retrieval & Evaluation.

Shared dataset structure:

```
`
memory_provider_eval/
├── README.md
├── memory_provider_comparison.ipynb
├── deterministic/
│ ├── spec.json
│ ├── documents/ (15 .md files)
│ └── ground_truth_qa.json
└── non_deterministic/
├── index.json
├── persona-01_priya.json ... persona-11_ravi.json
└── README.md
`
```

2.a Loading shared dataset

In [5]:

```
import glob
import json
import os
from pathlib import Path


#replace with own path to data root
DATA_ROOT = Path.cwd()

DET_DIR = os.path.join(DATA_ROOT, "deterministic")
DET_DOC_PATHS = sorted(glob.glob(os.path.join(DET_DIR, "documents",
"*.md")))
DET_DOCS = []
for p in DET_DOC_PATHS:
    with open(p) as f:
        DET_DOCS.append({"filename": os.path.basename(p), "text":
f.read()})

with open(os.path.join(DET_DIR, "ground_truth_qa.json")) as f:
    DET_QA = json.load(f)

ND_DIR = os.path.join(DATA_ROOT, "non_deterministic")
ND_PATHS = sorted(glob.glob(os.path.join(ND_DIR, "persona-*.json")))
ND_DOCS = []
for p in ND_PATHS:
    with open(p) as f:
        ND_DOCS.append(json.load(f))

print(f"Loaded {len(DET_DOCS)} deterministic documents, {len(DET_QA)}
deterministic QA pairs")
print(f"Loaded {len(ND_DOCS)} custom contradiction personas")
```

```
Loaded 15 deterministic documents, 29 deterministic QA pairs
Loaded 11 custom contradiction personas
```

2.b Ingestion for each provider

In this order:
1. Zep
2. mem0
3. Supermemory
5. Cognee

In [6]:

```
from zep_cloud.client import Zep
from zep_cloud.types import Message
from zep_cloud.core.api_error import ApiError

zep_client = Zep(api_key=os.environ["ZEP_API_KEY"])

def _ensure_user(user_id, first_name, last_name):
    try:
        zep_client.user.add(user_id=user_id, first_name=first_name,
last_name=last_name)
    except ApiError as e:
        if "already exists" not in str(e.body):
            raise
def _role_for_speaker(speaker, persona_name):
    return "user" if speaker == persona_name else "assistant"

def _ensure_thread(thread_id, user_id):
    try:
        zep_client.thread.create(thread_id=thread_id, user_id=user_id)
    except ApiError as e:
        if "already exists" not in str(e.body):
            raise

def zep_ingest_documents():
    _ensure_user("solvane_docs", "Solvane", "Docs")
    for doc in DET_DOCS:
        zep_client.graph.add(user_id="solvane_docs", type="text",
data=doc["text"])

def zep_ingest_conversations():
    for persona in ND_DOCS:
        user_id = persona["persona_id"]
        persona_name = persona["persona_name"]
        _ensure_user(user_id, persona_name.split()[0], "Persona")
        for session in persona["sessions"]:
            thread_id = f"
{persona['persona_id']}_{session['session_id']}"
            _ensure_thread(thread_id, user_id)
            messages = [
                Message(
                    name=t["speaker"],
                    content=t["text"],
                    role=_role_for_speaker(t["speaker"],
persona_name),
                )
                for t in session["turns"]
                if t["text"] != "PLACEHOLDER_RESPONSE_TO_EVALUATE"
            ]
            if messages:
                zep_client.thread.add_messages(thread_id=thread_id,
messages=messages)

zep_ingest_documents()
zep_ingest_conversations()

print(f"Zep ingestion complete: {len(DET_DOCS)} documents and
{len(ND_DOCS)} personas loaded")
```

```
Zep ingestion complete: 15 documents and 11 personas loaded
```

In [7]:

```
from mem0 import MemoryClient
import os
from dotenv import load_dotenv

load_dotenv()
mem0_client = MemoryClient(api_key=os.getenv("MEM0_API_KEY"))

def mem0_ingest_documents():
    for doc in DET_DOCS:
        mem0_client.add(
            user_id="solvane_docs",
            run_id="solvane_docs_thread",
            messages=[{"role": "system", "content": doc["text"]}],
        )

def mem0_ingest_conversations():
    for persona in ND_DOCS:
        user_id = persona["persona_id"]
        for session in persona["sessions"]:
            run_id = f"
{persona['persona_id']}_{session['session_id']}"
            messages = [
                {"role": t["speaker"], "content": t["text"]}
                for t in session["turns"] if t["text"] !=
"PLACEHOLDER_RESPONSE_TO_EVALUATE"
            ]
            if messages:
                mem0_client.add(user_id=user_id, run_id=run_id,
messages=messages)

mem0_ingest_documents()
mem0_ingest_conversations()

print(f"mem0 ingestion complete: {len(DET_DOCS)} documents and
{len(ND_DOCS)} personas loaded")
```

```
mem0 ingestion complete: 15 documents and 11 personas loaded
```

In [8]:

```
from supermemory import Supermemory
import os
from dotenv import load_dotenv

load_dotenv()
sm_client = Supermemory(api_key=os.getenv("SUPERMEMORY_API_KEY"))

def supermemory_ingest_documents():
    for doc in DET_DOCS:
        sm_client.add(content=doc["text"],
container_tag="solvane_docs")

def supermemory_ingest_conversations():
    for persona in ND_DOCS:
        tag = persona["persona_id"]
        for session in persona["sessions"]:
            for t in session["turns"]:
                if t["text"] == "PLACEHOLDER_RESPONSE_TO_EVALUATE":
                    continue
                sm_client.add(content=f"{t['speaker']}: {t['text']}",
container_tag=tag)

supermemory_ingest_documents()
supermemory_ingest_conversations()

print(f"Supermemory ingestion complete: {len(DET_DOCS)} documents and
{len(ND_DOCS)} personas loaded")
```

```
Supermemory ingestion complete: 15 documents and 11 personas
loaded
```

In [ ]:

```
import os
import cognee

async def cognee_ingest_documents():
    for doc in DET_DOCS:
        await cognee.remember(
            f"[{doc['filename']}]\n{doc['text']}",
            dataset_name="solvane_docs",
        )

async def cognee_ingest_conversations():
    for persona in ND_DOCS:
        for session in persona["sessions"]:
            for turn in session["turns"]:
                if turn["text"] == "PLACEHOLDER_RESPONSE_TO_EVALUATE":
                    continue
                await cognee.remember(
                    f"[{persona['persona_id']} /
{session['session_id']} / {session['date']}] "
                    f"{turn['speaker']}: {turn['text']}",
                    dataset_name=persona["persona_id"],
                )

await cognee.serve(
    url=os.getenv("COGNEE_URL"),
    api_key=os.getenv("COGNEE_API_KEY"),
)

await cognee_ingest_documents()
await cognee_ingest_conversations()

print(f"Cognee ingestion complete: {len(DET_DOCS)} documents and
{len(ND_DOCS)} personas loaded")
```

Confirm data ingestion actually landed and is queryable:

Several of these providers build their memory representation (graph edges, embeddings, indices) asynchronously after an add/remember call returns, so the call finishing without an error doesn't guarantee the data is queryable yet. Querying too soon after ingestion is a plausible source of run-to-run variance in the results below (a provider that hasn't finished indexing looks worse than one queried a minute later, even on identical data, independent of any LLM temperature setting). The cell below polls each provider a few times with a short wait between attempts, rather than checking once and moving on.

In [10]:

```
import time
import asyncio

_CHECK_PERSONA = ND_DOCS[0]["persona_id"]  # representative sample,
not exhaustive over all 11

def _poll_until_nonzero(label, check_fn, attempts=4, delay=5):
    for attempt in range(attempts):
        try:
            count = check_fn()
        except Exception as exc:
            count = None
            print(f"  {label}: ERROR on attempt {attempt +
1}/{attempts} -- {exc}")
        else:
            if count and count > 0:
                suffix = "" if attempt == 0 else f" (took {attempt +
1} attempts)"
                print(f"  {label}: OK -- queryable{suffix}")
                return True
            print(f"  {label}: not yet queryable (attempt {attempt +
1}/{attempts}), retrying...")
        if attempt < attempts - 1:
            time.sleep(delay)
    print(f"  {label}: WARNING -- still not queryable after {attempts}
attempts x {delay}s. "
          f"Ingestion may not have finished indexing, or didn't
actually land.")
    return False

async def _poll_until_nonzero_async(label, check_fn, attempts=4,
delay=5):
    for attempt in range(attempts):
        try:
            count = await check_fn()
        except Exception as exc:
            count = None
            print(f"  {label}: ERROR on attempt {attempt +
1}/{attempts} -- {exc}")
        else:
            if count and count > 0:
                suffix = "" if attempt == 0 else f" (took {attempt +
1} attempts)"
                print(f"  {label}: OK -- queryable{suffix}")
                return True
            print(f"  {label}: not yet queryable (attempt {attempt +
1}/{attempts}), retrying...")
        if attempt < attempts - 1:
            await asyncio.sleep(delay)
    print(f"  {label}: WARNING -- still not queryable after {attempts}
attempts x {delay}s. "
          f"Ingestion may not have finished indexing, or didn't
actually land.")
    return False

print("--- Zep ---")
for uid in ["solvane_docs", _CHECK_PERSONA]:
    _poll_until_nonzero(uid, lambda uid=uid: len(
        zep_client.graph.search(user_id=uid, query="summary",
scope="edges", limit=3).edges
    ))

print("--- mem0 ---")
for uid in ["solvane_docs", _CHECK_PERSONA]:
    _poll_until_nonzero(uid, lambda uid=uid: len(
        mem0_client.search("What do you know so far?", filters=
{"user_id": uid})["results"]
    ))

print("--- Supermemory ---")
for tag in ["solvane_docs", _CHECK_PERSONA]:
    _poll_until_nonzero(tag, lambda tag=tag: len(
        sm_client.search.memories(q=tag, container_tag=tag,
search_mode="hybrid", threshold=0, limit=5).results
    ))

print("--- Cognee ---")
async def _cognee_count(dataset_name):
    results = await cognee.recall(query_text="summary", datasets=
[dataset_name], only_context=True)
    if isinstance(results, str):
        return 1 if results.strip() else 0
    return len(results) if results else 0

for dataset_name in ["solvane_docs", _CHECK_PERSONA]:
    await _poll_until_nonzero_async(dataset_name, lambda
dataset_name=dataset_name: _cognee_count(dataset_name))
```

```
--- Zep ---
  solvane_docs: OK -- queryable
  persona-01: OK -- queryable
--- mem0 ---
  solvane_docs: OK -- queryable
  persona-01: OK -- queryable
--- Supermemory ---
  solvane_docs: OK -- queryable
  persona-01: OK -- queryable
--- Cognee ---
  solvane_docs: OK -- queryable
  persona-01: OK -- queryable
```

3. Retrieval & Eval

With data ingested across all four providers, we query each system with three independent evaluation sets: deterministic ground-truth QA, ambiguous/contradictory persona probes, and needle-in-haystack recall. The per-provider query functions defined below (`zep_ask`, `mem0_ask`, `supermemory_ask`, `cognee_ask`) are shared across all three: they route every answer through one shared `_synthesize` call, switched between a "resolve" and "hedge" system prompt depending on which section is calling it (see note below).

#### a. Ground-Truth QA

Questions from `ground_truth_qa.json`, each with one defined correct answer. Automatically split by type into: plain factual recall, temporal reasoning (facts that change over time), conflict resolution (superseded facts as of a given date), and multi-hop reasoning.

Retrieval + synthesis, shared across sections (a)-(c)

Each `*_ask` function wraps that provider's native retrieval call, then routes the result through `_synthesize`. This synthesis step is what's actually graded, not raw memory output, so all four providers are judged on the same basis.

`_synthesize` runs in one of two modes, toggled by the global `_SYNTHESIS_MODE`:
- **resolve** (default, used in sections a and c): instructs the model to commit to one answer, picking the fact valid at the target date if several conflict. Every question here has one correct answer, so hedging would itself be a wrong answer.
- **hedge** (set only inside `evaluate_ambiguity` / `evaluate_ambiguity_async`, section b): instructs the model to surface unresolved contradictions instead of silently picking a side.

In [11]:

```
import os
from openai import OpenAI
import re
import time


_llm = OpenAI(api_key=os.environ["OPENAI_API_KEY"])
_DATE_CLAUSE_RE = re.compile(r"\s*as of \d{4}-\d{2}-\d{2}\??",
re.IGNORECASE)

# _SYNTHESIS_MODE controls which system prompt _synthesize uses
_SYNTHESIS_MODE = "resolve"

_RESOLVE_SYSTEM_PROMPT = (
    "Answer using ONLY the facts listed below. Facts may include a
validity range "
    "like '(valid: DATE to DATE)' where 'present/current' means still
true. Ignore "
    "today's real-world date. If facts conflict, pick the one valid at
the target date. "
    "Be concise: state just the answer.\n\n"
    "Special case: if the question asks what a specific past document,
snapshot, or "
    "report stated (e.g. 'according to the April 2024 org
snapshot...') AND whether that "
    "is 'still accurate' or 'still true', answer BOTH parts: (1) what
the fact was at "
    "that past point in time, and (2) whether it has since been
superseded -- citing the "
    "newer fact if so. Do not just give the current state; the
question is explicitly "
    "asking you to compare the historical claim against the current
one."
)

_HEDGE_SYSTEM_PROMPT = (
    "Answer using ONLY the facts listed below. Some of these facts may
be genuinely "
    "ambiguous or contradictory with no resolution given -- do not
invent one. If the facts "
    "conflict and nothing in them tells you which one is current or
correct, say so "
    "explicitly and present both sides rather than confidently picking
one. If the facts "
    "are not actually in conflict, or an explicit resolution IS given
(e.g. a later fact "
    "that plainly supersedes an earlier one), answer directly and do
not hedge unnecessarily."
)

# --- Retry-with-backoff for every external network call -------------
------
# This notebook makes a very large number of sequential live API calls
(OpenAI +
# whichever provider is under test). A single transient
timeout/connection error used to
# crash the entire run_and_log cell partway through, losing every
provider that hadn't run
# yet. _with_retries / _with_retries_async retry on
timeout/connection-shaped errors with
# exponential backoff (2s, 4s, 8s) before giving up. Tune
max_attempts/base_delay if your
# network or a specific provider is consistently slower than this.
def _is_retryable(exc):
    msg = str(exc).lower()
    return any(s in msg for s in ("timeout", "timed out",
"connection", "temporarily unavailable"))

def _with_retries(fn, *args, max_attempts=4, base_delay=2, **kwargs):
    last_exc = None
    for attempt in range(max_attempts):
        try:
            return fn(*args, **kwargs)
        except Exception as exc:
            last_exc = exc
            if not _is_retryable(exc) or attempt == max_attempts - 1:
                raise
            wait = base_delay * (2 ** attempt)
            print(f"  [retry] {type(exc).__name__}: {exc} -- attempt
{attempt + 1}/{max_attempts}, retrying in {wait}s")
            time.sleep(wait)
    raise last_exc

async def _with_retries_async(fn, *args, max_attempts=4, base_delay=2,
**kwargs):
    import asyncio
    last_exc = None
    for attempt in range(max_attempts):
        try:
            return await fn(*args, **kwargs)
        except Exception as exc:
            last_exc = exc
            if not _is_retryable(exc) or attempt == max_attempts - 1:
                raise
            wait = base_delay * (2 ** attempt)
            print(f"  [retry] {type(exc).__name__}: {exc} -- attempt
{attempt + 1}/{max_attempts}, retrying in {wait}s")
            await asyncio.sleep(wait)
    raise last_exc

def _strip_date_clause(question):
    return _DATE_CLAUSE_RE.sub("", question).strip()

def _synthesize(question, snippets, query_date=None):
    if isinstance(snippets, str):
        context = snippets or "(no facts retrieved)"
    else:
        context = "\n".join(f"- {s}" for s in snippets) if snippets
else "(no facts retrieved)"

    date_note = (f"\n\nAnswer as of: {query_date}." if query_date
                 else "\n\nNo date given -- answer with the
current/most recent state.")

    system_prompt = _HEDGE_SYSTEM_PROMPT if _SYNTHESIS_MODE == "hedge"
else _RESOLVE_SYSTEM_PROMPT

    resp = _with_retries(
        _llm.chat.completions.create,
        model="gpt-4o-mini",
        temperature=0,
        messages=[
            {"role": "system", "content": system_prompt},
            {"role": "user", "content":
f"Facts:\n{context}\n\nQuestion: {question}{date_note}"},
        ],
    )
    return resp.choices[0].message.content.strip()


def mem0_ask(question, query_date=None, user_id="solvane_docs"):
    search_query = _strip_date_clause(question)
    results = _with_retries(mem0_client.search, search_query, filters=
{"user_id": user_id})
    snippets = [r["memory"] for r in results["results"]]
    return _synthesize(question, snippets, query_date)

def zep_ask(question, query_date=None, user_id="solvane_docs"):
    search_query = _strip_date_clause(question)
    thread_id = f"eval-{uuid.uuid4().hex}"
    _with_retries(zep_client.thread.create, thread_id=thread_id,
user_id=user_id)
    _with_retries(
        zep_client.thread.add_messages,
        thread_id,
        messages=[Message(name="Evaluator", role="user",
content=search_query)],
    )

    ctx = _with_retries(zep_client.thread.get_user_context,
thread_id=thread_id)
    return _synthesize(question, ctx.context, query_date)


def supermemory_ask(question, query_date=None, tag="solvane_docs"):
    search_query = _strip_date_clause(question)
    r = _with_retries(
        sm_client.search.memories,
        q=search_query, container_tag=tag, search_mode="hybrid",
threshold=0, limit=10,
    )
    snippets = [res.memory or res.chunk for res in r.results]
    return _synthesize(question, snippets, query_date)
```

#### Grading utilities

`ask_safe`: catches provider errors so one failure doesn't kill the run

`llm_judge`: grades free-text answers when they don't match the reference verbatim

`_contains_whole`: word-boundary substring check used before falling back to `llm_judge`. A plain `in` check would let a wrong answer like "75 kg" pass when the reference is "5 kg", since the shorter string sits inside the longer one

`CATEGORY_BY_TYPE`: buckets each ground-truth question into Temporal / Conflict Resolution / Multi-hop

These four are reused in section (c); `evaluate_provider` below is the only part specific to ground-truth QA.

Limitation: the same model family (gpt-4o-mini) both answers questions (via `_synthesize`) and grades them (via `llm_judge` / `annotation_judge`). That's a self-grading setup, see section (f) for a manual spot-check sample to validate the judge isn't rubber-stamping itself.

In [12]:

```
from collections import defaultdict
import re

CATEGORY_BY_TYPE = {
    "as_of": "Temporal Reasoning",
    "current_undated": "Conflict Resolution",
    "multi_hop": "Multi-hop Reasoning",
    "distractor_check": "Multi-hop Reasoning",
}

def normalize_answer(text):
    return " ".join(str(text or "").strip().lower().split())

def _contains_whole(needle, haystack):
    """Word-boundary substring check. Plain `in` would match '5 kg'
inside '75 kg' or '15 kg';
    this requires the needle to sit on a word boundary on both
sides."""
    if not needle:
        return False
    pattern = r"(?<!\w)" + re.escape(needle) + r"(?!\w)"
    return re.search(pattern, haystack) is not None

def ask_safe(question, query_date, ask_fn, *extra_args):
    try:
        return str(ask_fn(question, query_date, *extra_args))
    except Exception as exc:
        return f"<ERROR: {exc}>"

async def ask_safe_async(question, query_date, ask_fn, *extra_args):
    try:
        return str(await ask_fn(question, query_date, *extra_args))
    except Exception as exc:
        return f"<ERROR: {exc}>"

def llm_judge(question, expected, got):
    try:
        resp = _with_retries(
            _llm.chat.completions.create,
            model="gpt-4o-mini",
            temperature=0,
            messages=[
                {"role": "system", "content": (
                    "Grade a candidate answer against a reference
answer. Mark CORRECT if the "
```

```
                    "candidate's core factual claim matches the
reference — even if phrased very "
                    "differently, more verbosely, with extra true
context, or using equivalent "
                    "wording/units (e.g. '$150 per seat per month' =
'$150/seat/month'; 'Tuesday to "
                    "Thursday' = 'Tue/Wed/Thu').\n"
                    "Do not penalize: paraphrase, extra correct
detail, or additional caveats/exceptions "
                    "that coexist with the core claim without
contradicting it (e.g. if the reference says "
                    "'3 days/week in office (Tue/Wed/Thu)' and the
candidate adds 'Monday and Friday are "
                    "remote-flexible, with full-remote by manager
exception,' that is CORRECT — it doesn't "
                    "change the number of in-office days or which days
they are).\n"
                    "Mark INCORRECT only if the candidate: states a
different fact/value than the "
                    "reference, changes the specific
number/days/amount the reference commits to, omits "
                    "part of a multi-part reference answer, actually
contradicts the reference, or claims "
                    "no information when the reference has a specific
answer.\n"
                    "Respond in exactly this format:\n"
                    "VERDICT: CORRECT or INCORRECT\n"
                    "REASON: one short sentence"
                )},
                {"role": "user", "content": f"Question:
{question}\nReference answer: {expected}\nCandidate answer: {got}"},
            ],
        )
    except Exception as exc:
        # Retries exhausted -- don't crash the whole evaluate_provider
loop over one
        # stubborn timeout. Marked as a distinguishable FAIL so it's
easy to filter out
        # of the accuracy numbers rather than silently counting as a
real wrong answer.
        return False, f"<JUDGE_ERROR: {exc}>"

    text = resp.choices[0].message.content.strip()
    verdict_line = next((l for l in text.splitlines() if
l.upper().startswith("VERDICT")), "")
    reason_line = next((l for l in text.splitlines() if
l.upper().startswith("REASON")), "")
    is_correct = "CORRECT" in verdict_line.upper() and "INCORRECT" not
in verdict_line.upper()
    return is_correct, reason_line.split(":", 1)[-1].strip()

def qa_category(qa):
    return CATEGORY_BY_TYPE.get(qa["type"], "Other")


def evaluate_provider(name, ask_fn):
    correct = 0
    cat_correct, cat_total = defaultdict(int), defaultdict(int)

    for qa in DET_QA:
        question, query_date = qa["question"], qa.get("query_date")
        display_q = f"{question} (Answer as of {query_date})" if
query_date else question

        answer = ask_safe(question, query_date, ask_fn)
        norm_answer, norm_expected = normalize_answer(answer),
normalize_answer(qa["expected_answer"])

        if norm_answer == norm_expected or
_contains_whole(norm_expected, norm_answer):
            is_correct, reason = True, "exact/substring match"
        else:
            is_correct, reason = llm_judge(display_q,
qa["expected_answer"], answer)

        status = "PASS" if is_correct else "FAIL"
        correct += is_correct
        category = qa_category(qa)
        cat_total[category] += 1
        cat_correct[category] += is_correct

        print(f"{name} {qa['id']} [{category}] {status} — {reason}")
        print(" Q:", display_q)
        print(" expected:", qa["expected_answer"])
        print(" got     :", answer)
        print()

    print(f"{name} overall: {correct}/{len(DET_QA)}
({correct/len(DET_QA):.1%})")
    for category in ["Temporal Reasoning", "Conflict Resolution",
"Multi-hop Reasoning"]:
        c, t = cat_correct[category], cat_total[category]
        if t:
            print(f"  {category}: {c}/{t} ({c/t:.1%})")

    return {
        "overall": (correct, len(DET_QA)),
        "by_category": {cat: (cat_correct[cat], cat_total[cat]) for
cat in cat_total},
    }
```

Cognee's retrieval and eval loop are defined separately since its SDK is async-only. `cognee_ask` and `evaluate_provider_async` mirror the synchronous versions above one-for-one.

In [13]:

```
import cognee

async def cognee_ask(question, query_date=None,
dataset_name="solvane_docs"):
    search_query = _strip_date_clause(question)

    results = await _with_retries_async(
        cognee.recall,
        query_text=search_query,
        datasets=[dataset_name],
        only_context=True,
    )

    if isinstance(results, str):
        context = results
    elif results and isinstance(results[0], dict):
        context = "\n\n".join(r.get("text", "") for r in results if
r.get("text"))
    else:
        context = "\n".join(str(r) for r in results) if results else
""

    return _synthesize(question, context, query_date)


async def evaluate_provider_async(name, ask_fn):
    correct = 0
    cat_correct, cat_total = defaultdict(int), defaultdict(int)

    for qa in DET_QA:
        question, query_date = qa["question"], qa.get("query_date")
        display_q = f"{question} (Answer as of {query_date})" if
query_date else question

        answer = await ask_safe_async(question, query_date, ask_fn)
        norm_answer, norm_expected = normalize_answer(answer),
normalize_answer(qa["expected_answer"])

        if norm_answer == norm_expected or
_contains_whole(norm_expected, norm_answer):
            is_correct, reason = True, "exact/substring match"
        else:
            is_correct, reason = llm_judge(display_q,
qa["expected_answer"], answer)

        status = "PASS" if is_correct else "FAIL"
        correct += is_correct
        category = qa_category(qa)
        cat_total[category] += 1
        cat_correct[category] += is_correct

        print(f"{name} {qa['id']} [{category}] {status} — {reason}")
        print(" Q:", display_q)
        print(" expected:", qa["expected_answer"])
        print(" got     :", answer)
        print()

    print(f"{name} overall: {correct}/{len(DET_QA)}
({correct/len(DET_QA):.1%})")
    for category in ["Temporal Reasoning", "Conflict Resolution",
"Multi-hop Reasoning"]:
        c, t = cat_correct[category], cat_total[category]
        if t:
            print(f"  {category}: {c}/{t} ({c/t:.1%})")

    return {
        "overall": (correct, len(DET_QA)),
        "by_category": {cat: (cat_correct[cat], cat_total[cat]) for
cat in cat_total},
    }
```

#### b. Ambiguity & Contradiction Handling

Persona-based probes over genuinely contradictory user history, no single correct answer. Graded on whether the response hedges appropriately, silently picks a side, invents a resolution, or over-hedges when the conflict was actually resolved.

`evaluate_ambiguity` / `evaluate_ambiguity_async` switch `_synthesize` into "hedge" mode for the duration of their loop (see the note in section a). Without this, every provider will be graded on ambiguity while still being told by the shared prompt to pick a side, which would make the section unusable as a comparison signal.

What the four verdicts mean:

Every probe response gets classified into exactly one of these by `annotation_judge`, based on the specific tension documented for that persona (see each persona's `annotation` block):

- **GOOD** — the response surfaces the tension instead of silently picking a side (e.g. "you mentioned being vegetarian, but also had a ribeye recently — still keeping meals meat-free, or open to more now?"), *or*, for the two contrast-case personas where a real resolution was actually given (Morgan P.'s explicit budget correction, Ravi S.'s tentative deadline slip), correctly uses that resolution without hedging on something that isn't actually still open.
- **SILENT_PICK** — confidently answers with only one side of a genuine contradiction, with no acknowledgment that conflicting information exists. Example from an actual run: asked about Priya's diet, a provider answered "vegetarian" and never mentioned her ribeye dinner at all, thus treating an explicitly unresolved tension as settled fact.
- **HALLUCINATED_RESOLUTION** — invents an explanation or reconciliation the user never actually gave (e.g. "since you mentioned switching to pescatarian..." when the user said no such thing). Worse than SILENT_PICK, since it fabricates evidence rather than just omitting the conflict.
- **OVER_HEDGED** — only applies to Morgan P. and Ravi S. (the two personas where a resolution genuinely was given). Fires when a provider treats an already-resolved point as if it were still an open contradiction. Hedging is only a good behavior when there's an actual unresolved tension to hedge about.

One thing worth being explicit about for readers: "good" here does not mean "matches a fixed reference string" the way ground-truth QA does. Nine of the eleven personas are genuinely unresolved by design (see `non_deterministic/README.md`), so there is no single correct answer to match against. The judge is grading *behavior* (appropriately uncertain vs. silently confident vs. fabricated), not factual accuracy against a ground truth.

In [14]:

```
def annotation_judge(persona_name, probe_query, response, annotation):
    try:
        resp = _with_retries(
            _llm.chat.completions.create,
            model="gpt-4o-mini",
            temperature=0,
            messages=[
                {"role": "system", "content": (
                    "You are grading how an AI assistant handled a
genuinely ambiguous or contradictory "
                    "piece of user history. You are given the nature
of the tension, what a good response "
                    "looks like, and specific failure modes to watch
for. Classify the candidate response "
                    "into exactly one of:\n"
                    "GOOD — appropriately hedges, surfaces the
tension, or (for contrast cases) correctly "
                    "applies an actual explicit resolution without
over-hedging.\n"
                    "SILENT_PICK — confidently answers with one side
of the contradiction as if it were "
                    "settled, with no acknowledgment of the
conflicting information.\n"
                    "HALLUCINATED_RESOLUTION — invents an explanation
or reconciliation the user never gave.\n"
                    "OVER_HEDGED — treats something as ambiguous when
it was actually explicitly resolved "
                    "(only applies to contrast-case personas).\n"
                    "Respond in exactly this format:\n"
                    "VERDICT: GOOD or SILENT_PICK or
HALLUCINATED_RESOLUTION or OVER_HEDGED\n"
                    "REASON: one short sentence"
                )},
                {"role": "user", "content": (
                    f"Persona: {persona_name}\n"
                    f"Tension: {annotation['tension']}\n"
                    f"Good response looks like:
{annotation['good_response_looks_like']}\n"
                    f"Failure modes to watch for:
{annotation['failure_modes_to_watch_for']}\n\n"
                    f"Probe query: {probe_query}\n"
                    f"Candidate response: {response}"
                )},
            ],
        )
    except Exception as exc:
        # Retries exhausted -- degrade to UNKNOWN (a bucket the
results dict already
        # tracks) instead of crashing the whole evaluate_ambiguity
loop.
        return "UNKNOWN", f"<JUDGE_ERROR: {exc}>"

    text = resp.choices[0].message.content.strip()
    verdict_line = next((l for l in text.splitlines() if
l.upper().startswith("VERDICT")), "")
    reason_line = next((l for l in text.splitlines() if
l.upper().startswith("REASON")), "")
    verdict = verdict_line.split(":", 1)[-1].strip().upper() if ":" in
verdict_line else "UNKNOWN"
    reason = reason_line.split(":", 1)[-1].strip() if reason_line else
""
    return verdict, reason


def evaluate_ambiguity(name, ask_fn):
    global _SYNTHESIS_MODE
    results = {"GOOD": 0, "SILENT_PICK": 0, "HALLUCINATED_RESOLUTION":
0, "OVER_HEDGED": 0, "UNKNOWN": 0}
    _SYNTHESIS_MODE = "hedge"
    try:
        for persona in ND_DOCS:
            for probe in persona["probe_queries"]:
                response = ask_fn(probe, None, persona["persona_id"])
# sync providers; await for async ones
                verdict, reason =
annotation_judge(persona["persona_name"], probe, response,
persona["annotation"])
                results[verdict] = results.get(verdict, 0) + 1
                print(f"{name} {persona['persona_id']} [{verdict}] —
{reason}")
                print(" Probe:", probe)
                print(" Got:  ", response)
                print()
    finally:
        _SYNTHESIS_MODE = "resolve"

    total = sum(results.values())
    print(f"{name}: {results['GOOD']}/{total} good
({results['GOOD']/total:.1%}) — "
          f"{results['SILENT_PICK']} silent picks,
{results['HALLUCINATED_RESOLUTION']} hallucinated, "
          f"{results['OVER_HEDGED']} over-hedged")

async def evaluate_ambiguity_async(name, ask_fn):
    global _SYNTHESIS_MODE
    results = {"GOOD": 0, "SILENT_PICK": 0, "HALLUCINATED_RESOLUTION":
0, "OVER_HEDGED": 0, "UNKNOWN": 0}
    _SYNTHESIS_MODE = "hedge"
    try:
        for persona in ND_DOCS:
            for probe in persona["probe_queries"]:
                response = await ask_fn(probe, None,
persona["persona_id"])
                verdict, reason =
annotation_judge(persona["persona_name"], probe, response,
persona["annotation"])
                results[verdict] = results.get(verdict, 0) + 1
                print(f"{name} {persona['persona_id']} [{verdict}] —
{reason}")
                print(" Probe:", probe)
                print(" Got:  ", response)
                print()
    finally:
        _SYNTHESIS_MODE = "resolve"

    total = sum(results.values())
    print(f"{name}: {results['GOOD']}/{total} good
({results['GOOD']/total:.1%}) — "
          f"{results['SILENT_PICK']} silent picks,
{results['HALLUCINATED_RESOLUTION']} hallucinated, "
          f"{results['OVER_HEDGED']} over-hedged")
```

#### c. Needle in Haystack

Sparse, easily-missed facts buried in persona sessions. Tests recall precision rather than reasoning.

In [15]:

```
def evaluate_needles(name, ask_fn):
    correct = 0
    total = 0
    for persona in ND_DOCS:
        for probe in persona.get("needle_probes", []):
            total += 1
            question, expected = probe["question"],
probe["expected_answer"]
            answer = ask_safe(question, None, ask_fn,
persona["persona_id"])
            norm_answer, norm_expected = normalize_answer(answer),
normalize_answer(expected)

            if norm_answer == norm_expected or
_contains_whole(norm_expected, norm_answer):
                is_correct, reason = True, "exact/substring match"
            else:
                is_correct, reason = llm_judge(question, expected,
answer)

            status = "PASS" if is_correct else "FAIL"
            correct += is_correct
            print(f"{name} {persona['persona_id']} {status} —
{reason}")
            print(" Q:", question)
            print(" expected:", expected)
            print(" got     :", answer)
            print()

    if total:
        print(f"{name}: {correct}/{total} ({correct/total:.1%})")
    return correct, total


async def evaluate_needles_async(name, ask_fn):
    correct = 0
    total = 0
    for persona in ND_DOCS:
        for probe in persona.get("needle_probes", []):
            total += 1
            question, expected = probe["question"],
probe["expected_answer"]
            answer = await ask_safe_async(question, None, ask_fn,
persona["persona_id"])
            norm_answer, norm_expected = normalize_answer(answer),
normalize_answer(expected)

            if norm_answer == norm_expected or
_contains_whole(norm_expected, norm_answer):
                is_correct, reason = True, "exact/substring match"
            else:
                is_correct, reason = llm_judge(question, expected,
answer)

            status = "PASS" if is_correct else "FAIL"
            correct += is_correct
            print(f"{name} {persona['persona_id']} {status} —
{reason}")
            print(" Q:", question)
            print(" expected:", expected)
            print(" got     :", answer)
            print()

    if total:
        print(f"{name}: {correct}/{total} ({correct/total:.1%})")
    return correct, total
```

#### d. Run Evaluations

This is where retrieval actually happens: each provider is queried against all three evaluation sets above, and results are logged for comparison below.

In [ ]:

```
%pip install matplotlib pandas
import sys, io, re, contextlib
import pandas as pd
import matplotlib.pyplot as plt

class _Tee(io.StringIO):
    def __init__(self, real_stream):
        super().__init__()
        self._real = real_stream
    def write(self, s):
        self._real.write(s)
        return super().write(s)
    def flush(self):
        self._real.flush()

@contextlib.contextmanager
def tee_stdout():
    tee = _Tee(sys.stdout)
    with contextlib.redirect_stdout(tee):
        yield tee

# DET_QA / needle-in-haystack lines look like: "Provider id [Category]
PASS — reason"
PASS_FAIL_RE = re.compile(r"^(\S+)\s+(\S+)\s+(?:\[(.+?)\]\s+)?
(PASS|FAIL)\b")
# Ambiguity lines look like: "Provider persona-id
[GOOD/SILENT_PICK/...] — reason"
AMBIGUITY_RE = re.compile(r"^(\S+)\s+(\S+)\s+\
[(GOOD|SILENT_PICK|HALLUCINATED_RESOLUTION|OVER_HEDGED|UNKNOWN)\]")

results_log = []
# raw_logs keeps the full printed text (question/expected/got/reason)
per run, not just the
# parsed PASS/FAIL booleans in results_log — needed so section (f) can
sample real judged
# answers for a human spot-check instead of just re-trusting the
aggregate pass rate
raw_logs = []

def _capture_to_log(raw_text, default_category=None):
    for line in raw_text.splitlines():
        line = line.strip()
        m = PASS_FAIL_RE.match(line)
        if m:
            provider, qid, category, status = m.groups()
            results_log.append({
                "provider": provider, "id": qid,
                "category": category or default_category,
                "correct": status == "PASS",
            })
            continue
        m = AMBIGUITY_RE.match(line)
        if m:
            provider, qid, verdict = m.groups()
            results_log.append({
                "provider": provider, "id": qid,
                "category": default_category or "Ambiguity",
                "correct": verdict == "GOOD",
            })

def run_and_log(name, ask_fn, eval_fn, category=None):
    with tee_stdout() as buf:
        eval_fn(name, ask_fn)
    text = buf.getvalue()
    _capture_to_log(text, default_category=category)
    raw_logs.append((name, category or "Ground-Truth QA", text))

async def run_and_log_async(name, ask_fn, eval_fn, category=None):
    with tee_stdout() as buf:
        await eval_fn(name, ask_fn)
    text = buf.getvalue()
    _capture_to_log(text, default_category=category)
    raw_logs.append((name, category or "Ground-Truth QA", text))
```

In [17]:

```
# DET_QA — category comes from the [Temporal Reasoning]/[Conflict
Resolution]/[Multi-hop] tag automatically
run_and_log("Zep", zep_ask, evaluate_provider)
run_and_log("Mem0", mem0_ask, evaluate_provider)
run_and_log("Supermemory", supermemory_ask, evaluate_provider)
await run_and_log_async("Cognee", cognee_ask, evaluate_provider_async)

# Needle-in-haystack
run_and_log("Zep", zep_ask, evaluate_needles, category="Needle-in-
Haystack")
run_and_log("Mem0", mem0_ask, evaluate_needles, category="Needle-in-
Haystack")
run_and_log("Supermemory", supermemory_ask, evaluate_needles,
category="Needle-in-Haystack")
await run_and_log_async("Cognee", cognee_ask, evaluate_needles_async,
category="Needle-in-Haystack")

# Ambiguity (non-det)
run_and_log("Zep", zep_ask, evaluate_ambiguity, category="Ambiguity")
run_and_log("Mem0", mem0_ask, evaluate_ambiguity,
category="Ambiguity")
run_and_log("Supermemory", supermemory_ask, evaluate_ambiguity,
category="Ambiguity")
await run_and_log_async("Cognee", cognee_ask,
evaluate_ambiguity_async, category="Ambiguity")
```

```
Zep det-001 [Temporal Reasoning] PASS — exact/substring match
 Q: Who leads Team Nova as of 2024-06-01? (Answer as of 2024-06-
01)
 expected: Alice Chen
 got     : As of June 1, 2024, Team Nova is led by Alice Chen.
This information is still accurate and has not been superseded.

Zep det-002 [Temporal Reasoning] FAIL — The candidate states that
Alice Chen leads Team Nova, which contradicts the reference
answer that states Bob Okafor is the leader.
 Q: Who leads Team Nova as of 2025-06-01? (Answer as of 2025-06-
01)
 expected: Bob Okafor
 got     : As of June 1, 2025, Team Nova is led by Alice Chen.
This information is still accurate and has not been superseded
since the April 2024 org snapshot.

Zep det-003 [Conflict Resolution] FAIL — The candidate states a
different leader than the reference.
 Q: Who leads Team Nova?
 expected: Bob Okafor
 got     : Alice Chen leads Team Nova.

Zep det-004 [Temporal Reasoning] PASS — exact/substring match
 Q: Who leads Team Forge as of 2024-03-01? (Answer as of 2024-03-
01)
 expected: Sofia Kappler
 got     : As of March 1, 2024, Team Forge is led by Sofia
Kappler. This information is still accurate as of that date.

Zep det-005 [Temporal Reasoning] FAIL — The candidate states that
Sofia Kappler leads Team Forge, which contradicts the reference
answer that names Derek Osei as the leader.
 Q: Who leads Team Forge as of 2025-01-01? (Answer as of 2025-01-
01)
 expected: Derek Osei
 got     : As of January 1, 2025, Team Forge is led by Sofia
Kappler. This is based on the April 2024 org snapshot, which
stated that Sofia Kappler was the lead of Team Forge at that
time. Since then, there have been no updates indicating a change
in leadership prior to January 1, 2025.

Zep det-006 [Conflict Resolution] PASS — exact/substring match
 Q: Who leads Team Forge?
 expected: Derek Osei
 got     : Derek Osei leads Team Forge.

Zep det-007 [Temporal Reasoning] PASS — exact/substring match
 Q: What is the maximum payload of the Solvane S1 as of 2023-06-
01? (Answer as of 2023-06-01)
 expected: 5 kg
 got     : As of June 1, 2023, the maximum payload of the Solvane
S1 was rated at 5 kg according to the original v1.0 spec sheet.
This rating has not been superseded by any newer information at
that time.

Zep det-008 [Temporal Reasoning] PASS — exact/substring match
 Q: What is the maximum payload of the Solvane S1 as of 2024-08-
01? (Answer as of 2024-08-01)
 expected: 7 kg
 got     : As of August 1, 2024, the maximum payload of the
Solvane S1 was 7 kg according to the v2.0 spec sheet published on
June 3, 2024. However, this figure was later corrected to 6 kg on
November 20, 2024, due to field reports of actuator strain.
Therefore, the 7 kg figure was accurate at that time but would be
superseded by the 6 kg rating after the correction.

Zep det-009 [Temporal Reasoning] PASS — exact/substring match
 Q: What is the maximum payload of the Solvane S1 as of 2025-06-
01? (Answer as of 2025-06-01)
 expected: 6 kg
 got     : As of June 1, 2025, the maximum payload of the Solvane
S1 was 7 kg according to the v2.0 spec sheet published on June 3,
2024. However, this figure was later corrected to 6 kg on
November 20, 2024, due to field reports of actuator strain.
Therefore, the 7 kg figure was not accurate as of June 1, 2025,
and the real-world validated maximum payload for S1 v2 hardware
is 6 kg.

Zep det-010 [Conflict Resolution] PASS — exact/substring match
 Q: What is the maximum payload of the Solvane S1?
 expected: 6 kg
 got     : The maximum payload of the Solvane S1 is 6 kg, as
corrected from the previous 7 kg figure.

Zep det-011 [Temporal Reasoning] FAIL — The candidate states that
the pricing is not specified, which contradicts the reference
answer.
 Q: What is the pricing for Solvane Vision Suite as of 2024-01-
01? (Answer as of 2024-01-01)
 expected: $499/month flat, individual license
 got     : As of January 1, 2024, the pricing for Solvane Vision
Suite is not specified in the provided facts. The information
regarding the transition to per-seat pricing at $150/seat/month
was posted on January 15, 2025, and therefore is not applicable
to the date in question.

Zep det-012 [Temporal Reasoning] PASS — The candidate's core
factual claim about the pricing matches the reference answer.
 Q: What is the pricing for Solvane Vision Suite as of 2025-06-
01? (Answer as of 2025-06-01)
 expected: $150/seat/month, per-seat license
 got     : As of June 1, 2025, the pricing for Solvane Vision
Suite is $150 per seat per month, following the transition from
the flat-rate individual tier of $499 per month, which was
discontinued on January 15, 2025.

Zep det-013 [Conflict Resolution] PASS — The core factual claim
about the pricing of $150 per seat per month matches the
reference answer.
 Q: What is the pricing for Solvane Vision Suite?
 expected: $150/seat/month, per-seat license (individual tier
discontinued)
 got     : The pricing for Solvane Vision Suite is $150 per seat
per month, effective January 15, 2025. This change has
discontinued the previous flat-rate individual tier of $499 per
month.

Zep det-014 [Temporal Reasoning] FAIL — The candidate states that
the remote work policy as of January 1, 2024, is still based on a
fully remote structure, which contradicts the reference answer.
 Q: What is Solvane's remote work policy as of 2024-01-01?
(Answer as of 2024-01-01)
 expected: Fully remote, no in-office requirement
 got     : As of January 1, 2023, Solvane Robotics had a fully
remote work policy, allowing teams to meet in person at their
discretion. However, starting February 1, 2025, employees,
including Solvane Docs, will be required to work from an assigned
Solvane office three days per week (Tuesday, Wednesday,
Thursday), with Mondays and Fridays remaining remote-flexible.
Therefore, the remote work policy as of January 1, 2024, is still
based on the fully remote structure, but it will change in the
future.

Zep det-015 [Temporal Reasoning] PASS — The candidate accurately
describes the core aspect of the remote work policy, which is 3
days in the office on specified days.
 Q: What is Solvane's remote work policy as of 2025-06-01?
(Answer as of 2025-06-01)
 expected: Hybrid: 3 days/week in office (Tue/Wed/Thu)
 got     : As of June 1, 2025, Solvane Robotics has a remote work
policy that requires employees to work from an assigned Solvane
office three days per week (Tuesday, Wednesday, Thursday), with
Mondays and Fridays remaining remote-flexible. This policy
supersedes the previous fully remote work policy.

Zep det-016 [Conflict Resolution] FAIL — The candidate states
that Solvane is fully remote, which contradicts the reference's
claim of a hybrid model requiring in-office work.
 Q: What is Solvane's remote work policy?
 expected: Hybrid: 3 days/week in office (Tue/Wed/Thu)
 got     : Solvane Robotics is a fully remote company that
designs warehouse automation hardware and software, with no
offices or required work locations. Teams can meet in person at
their discretion. Starting February 1, 2025, employees are
required to work from an assigned Solvane office three days per
week (Tuesday, Wednesday, Thursday), while Mondays and Fridays
remain remote-flexible.

Zep det-017 [Temporal Reasoning] PASS — exact/substring match
 Q: What is the daily meal expense reimbursement cap as of 2024-
01-01? (Answer as of 2024-01-01)
 expected: $50/day
 got     : The daily meal expense reimbursement cap as of January
1, 2024, is $50/day. This cap was later raised to $75/day
effective August 1, 2025.

Zep det-018 [Temporal Reasoning] PASS — exact/substring match
 Q: What is the daily meal expense reimbursement cap as of 2025-
12-01? (Answer as of 2025-12-01)
 expected: $75/day
 got     : The daily meal expense reimbursement cap as of
December 1, 2025, is $75/day. This cap was raised from $50/day
effective August 1, 2025, and receipts are required for any
single expense over $25.

Zep det-019 [Conflict Resolution] PASS — exact/substring match
 Q: What is the daily meal expense reimbursement cap?
 expected: $75/day
 got     : The daily meal expense reimbursement cap is $75/day
for business travel, effective August 1, 2025. Receipts are
required for single expenses over $25, and all other terms from
Handbook Section 7 remain unchanged.

Zep det-020 [Temporal Reasoning] PASS — The candidate's answer
matches the reference answer regarding battery life in both
modes.
 Q: What is the battery life of the Solvane Nav Kit as of 2025-
06-01? (Answer as of 2025-06-01)
 expected: 8 hours standard mode; up to 10 hours in eco mode
(both true, different modes)
 got     : The battery life of the Solvane Nav Kit is 8 hours in
standard mode and 10 hours in eco mode. This information is still
accurate as of June 1, 2025.

Zep det-021 [Conflict Resolution] PASS — The candidate accurately
states the battery life in both standard and eco modes as per the
reference.
 Q: What is the battery life of the Solvane Nav Kit?
 expected: Both values are current and true simultaneously (8h
standard mode; up to 10h eco mode) — there is no single 'most
recent supersedes all' answer here.
 got     : The Solvane Nav Kit has a battery life of 8 hours in
standard mode and up to 10 hours in eco mode.

Zep det-mh-001 [Multi-hop Reasoning] FAIL — The candidate
incorrectly states who reviewed the S1 payload issue and omits
the corrected payload rating.
 Q: Who reviewed the S1 payload issue that led to a leadership
change, and what is the corrected payload rating that came out of
it?
 expected: Derek Osei led the Team Forge review (becoming Team
Forge lead as a result, per the Sept 2024 incident report); the
corrected S1 v2 payload rating is 6 kg (per the Nov 2024 errata).
 got     : Solvane Docs reviewed the S1 payload issue that led to
a leadership change. The corrected payload rating resulting from
this review is not specified in the provided messages.

Zep det-mh-002 [Multi-hop Reasoning] PASS — The candidate
accurately states that Alice Chen is no longer the point of
contact and provides correct details about her transition and Bob
Okafor's succession.
 Q: As of mid-2025, is Alice Chen still the point of contact for
Team Nova? If not, why not and who is?
 expected: No — as of March 2025, Bob Okafor is Team Nova lead.
Alice Chen moved to a Principal Engineer (individual contributor)
role with no direct reports.
 got     : As of mid-2025, Alice Chen is not the point of contact
for Team Nova. She transitioned to a Principal Engineer role on
March 3, 2025, and Bob Okafor succeeded her as the point of
contact for Team Nova going forward.

Zep det-mh-003 [Multi-hop Reasoning] PASS — The candidate
accurately reflects the information from the reference about
Sofia Kappler being the lead in April 2024 and acknowledges the
change to Derek Osei.
 Q: According to the April 2024 org snapshot, who leads Team
Forge — and is that still accurate?
 expected: The April 2024 snapshot lists Sofia Kappler as Team
Forge lead, which was accurate at that time but is now stale —
Derek Osei became lead in September 2024.
 got     : According to the April 2024 org snapshot, Sofia
Kappler was identified as the leader of Team Forge. However, this
information is no longer accurate; Derek Osei is now the Lead of
Team Forge, effective as of the latest updates.

Zep det-mh-004 [Multi-hop Reasoning] FAIL — The candidate omits
the specific new roles and reasons for the transitions, which are
clearly stated in the reference.
 Q: Alice Chen and Sofia Kappler both stepped down as team leads.
What new roles did each move into, and what prompted each
transition?
 expected: Alice Chen moved to Principal Engineer, an individual-
contributor role with no direct reports, as part of a scheduled
leadership reorg (Bob Okafor succeeding her) -- not tied to any
incident. Sofia Kappler moved to a newly created Quality Systems
Advisor role, specifically prompted by the Team Forge review
following S1 payload field reports. The two transitions had
different causes: routine reorg for Alice, incident-driven review
for Sofia.
 got     : Alice Chen and Sofia Kappler both stepped down as team
leads, but the specific new roles they moved into and the reasons
prompting these transitions are not detailed.

Zep det-mh-005 [Multi-hop Reasoning] FAIL — The candidate states
that it is unknown, while the reference clearly states that the
correction does not apply to the original 2023 v1.0 hardware.
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 Q: Does the November 2024 S1 payload correction to 6 kg apply to
units running the original 2023 v1.0 hardware?
 expected: No. The errata explicitly corrects only the June 2024
v2.0 spec sheet's 7 kg figure; it does not affect the original
2023 v1.0 spec sheet, which remains rated at 5 kg.
 got     : It is unknown if the November 2024 S1 payload
correction to 6 kg applies to units running the original 2023
v1.0 hardware.

Zep det-mh-006 [Multi-hop Reasoning] FAIL — The candidate states
that Team Atlas received a new lead, contradicting the reference
that states there has been no leadership change for Team Atlas.
 Q: Team Nova and Team Forge both got new leads at different
points. Did Team Atlas ever get a new lead too?
 expected: No -- Marcus Webb has led Team Atlas continuously
since at least the 2023 handbook, and is still listed as Team
Atlas lead as recently as the October 2024 all-hands notes and
the April 2024 org snapshot. Unlike Nova and Forge, there is no
leadership change recorded for Team Atlas anywhere in the
dataset.
 got     : Team Atlas received a new lead as of July 24, 2026.

Zep det-mh-007 [Multi-hop Reasoning] FAIL — The candidate states
it is unclear if the announcement preceded the official notice,
which contradicts the reference.
 Q: Was the S1 v2 payload correction first made public at the
October 2024 all-hands, or was it announced there before the
official notice went out?
 expected: It was previewed internally at the October 22, 2024
all-hands (Derek walked through the correction ahead of the
public notice), but the official public errata wasn't posted
until November 20, 2024 -- about a month later.
 got     : The S1 v2 payload correction was previewed at the
October 2024 all-hands meeting, but it is unclear if this
announcement preceded the official notice issued on November 20,
2024.

Zep det-mh-008 [Multi-hop Reasoning] FAIL — The candidate's
answer incorrectly states that the information is still accurate
today, while the reference indicates that Alice Chen was the
first lead before Bob Okafor succeeded her.
 Q: The employee handbook's original org chart lists Team Nova's
lead as 'TBD.' Who ended up filling that role, and is that still
accurate today?
 expected: Alice Chen was announced as Team Nova's first lead in
January 2024, filling the 'TBD' slot from the 2023 handbook.
That's no longer current, though -- Bob Okafor succeeded her in
March 2025.
 got     : Bob Okafor filled the role of Lead for Team Nova
effective March 3, 2025. This is still accurate today.

Zep overall: 17/29 (58.6%)
  Temporal Reasoning: 10/14 (71.4%)
  Conflict Resolution: 5/7 (71.4%)
  Multi-hop Reasoning: 2/8 (25.0%)
Mem0 det-001 [Temporal Reasoning] PASS — exact/substring match
 Q: Who leads Team Nova as of 2024-06-01? (Answer as of 2024-06-
01)
 expected: Alice Chen
 got     : Alice Chen leads Team Nova as of June 1, 2024. This is
still accurate as of that date.

Mem0 det-002 [Temporal Reasoning] PASS — exact/substring match
 Q: Who leads Team Nova as of 2025-06-01? (Answer as of 2025-06-
01)
 expected: Bob Okafor
 got     : Bob Okafor leads Team Nova as of June 1, 2025.

Mem0 det-003 [Conflict Resolution] PASS — exact/substring match
 Q: Who leads Team Nova?
 expected: Bob Okafor
 got     : Bob Okafor leads Team Nova at Solvane Robotics.

Mem0 det-004 [Temporal Reasoning] PASS — exact/substring match
 Q: Who leads Team Forge as of 2024-03-01? (Answer as of 2024-03-
01)
 expected: Sofia Kappler
 got     : As of March 1, 2024, Sofia Kappler is the lead of Team
Forge. This is still accurate as of that date, as Derek Osei
became the lead effective September 16, 2024, which is after
March 1, 2024.

Mem0 det-005 [Temporal Reasoning] PASS — exact/substring match
 Q: Who leads Team Forge as of 2025-01-01? (Answer as of 2025-01-
01)
 expected: Derek Osei
 got     : As of January 1, 2025, Team Forge is led by Sofia
Kappler, who transitioned to a newly created Quality Systems
Advisor role on September 16, 2024. Therefore, Derek Osei has not
yet become the lead of Team Forge, as his effective date is
September 16, 2024.

Mem0 det-006 [Conflict Resolution] PASS — exact/substring match
 Q: Who leads Team Forge?
 expected: Derek Osei
 got     : Derek Osei leads Team Forge.

Mem0 det-007 [Temporal Reasoning] FAIL — The candidate states a
maximum payload of 6 kg, which differs from the reference answer
of 5 kg.
 Q: What is the maximum payload of the Solvane S1 as of 2023-06-
01? (Answer as of 2023-06-01)
 expected: 5 kg
 got     : The maximum payload of the Solvane S1 v2 hardware as
of June 1, 2023, is 6 kg. This is the validated and rated maximum
payload under real-world deployment conditions.

Mem0 det-008 [Temporal Reasoning] PASS — exact/substring match
 Q: What is the maximum payload of the Solvane S1 as of 2024-08-
01? (Answer as of 2024-08-01)
 expected: 7 kg
 got     : The maximum payload of the Solvane S1 as of 2024-08-01
is 6 kg. This reflects the updated rating following the errata
notice posted on November 20, 2024, which corrected the
previously published 7 kg figure from the June 2024 spec sheet.

Mem0 det-009 [Temporal Reasoning] PASS — exact/substring match
 Q: What is the maximum payload of the Solvane S1 as of 2025-06-
01? (Answer as of 2025-06-01)
 expected: 6 kg
 got     : The maximum payload of the Solvane S1 as of 2025-06-01
is 6 kg. This reflects the updated rating following the errata
notice posted on November 20, 2024, which corrected the previous
7 kg figure that was based on controlled lab conditions.

Mem0 det-010 [Conflict Resolution] PASS — exact/substring match
 Q: What is the maximum payload of the Solvane S1?
 expected: 6 kg
 got     : The maximum payload for the Solvane S1 v2 hardware is
6 kg.

Mem0 det-011 [Temporal Reasoning] PASS — The candidate's core
factual claim about the pricing of $499/month for an individual
license matches the reference answer.
 Q: What is the pricing for Solvane Vision Suite as of 2024-01-
01? (Answer as of 2024-01-01)
 expected: $499/month flat, individual license
 got     : As of January 1, 2024, the pricing for Solvane Vision
Suite is $499 per month for the Standard tier (flat rate,
individual license), covering a single deployment instance and
standard support. This pricing has been effective since May 1,
2023, and remains unchanged.

Mem0 det-012 [Temporal Reasoning] FAIL — The candidate states a
different price and licensing structure than the reference
answer.
 Q: What is the pricing for Solvane Vision Suite as of 2025-06-
01? (Answer as of 2025-06-01)
 expected: $150/seat/month, per-seat license
 got     : The pricing for Solvane Vision Suite as of June 1,
2025, is $499 per month for the Standard tier (flat rate,
individual license) covering a single deployment instance and
standard support. This pricing has been consistent since it was
effective on May 1, 2023.

Mem0 det-013 [Conflict Resolution] FAIL — The candidate states a
different pricing structure than the reference answer.
 Q: What is the pricing for Solvane Vision Suite?
 expected: $150/seat/month, per-seat license (individual tier
discontinued)
 got     : The pricing for Solvane Vision Suite is $499 per month
for the Standard tier (flat rate, individual license), covering a
single deployment instance and standard support. Enterprise
pricing is available on request for multi-site deployments.

Mem0 det-014 [Temporal Reasoning] FAIL — The candidate states
that the remote work policy was not specified, contradicting the
reference that states it is fully remote with no in-office
requirement.
 Q: What is Solvane's remote work policy as of 2024-01-01?
(Answer as of 2024-01-01)
 expected: Fully remote, no in-office requirement
 got     : As of January 1, 2024, Solvane Robotics had not yet
updated its remote work policy, which was still based on the 2023
edition of the employee handbook. The updated hybrid working
model, effective February 1, 2025, had not been implemented yet.
Therefore, the remote work policy prior to the update was not
specified in the provided facts.

Mem0 det-015 [Temporal Reasoning] PASS — The candidate accurately
describes the in-office days as Tuesday, Wednesday, and Thursday,
aligning with the reference answer.
 Q: What is Solvane's remote work policy as of 2025-06-01?
(Answer as of 2025-06-01)
 expected: Hybrid: 3 days/week in office (Tue/Wed/Thu)
 got     : As of June 1, 2025, Solvane Robotics' remote work
policy is that employees work from their assigned office on
Tuesday, Wednesday, and Thursday, while Monday and Friday remain
remote-flexible. Fully remote arrangements are still possible by
manager exception for roles hired as remote-only. This policy was
updated effective February 1, 2025, superseding the 2023 edition.

Mem0 det-016 [Conflict Resolution] PASS — The candidate
accurately describes the in-office days and adds additional
correct context about remote flexibility and exceptions.
 Q: What is Solvane's remote work policy?
 expected: Hybrid: 3 days/week in office (Tue/Wed/Thu)
 got     : Solvane Robotics has a hybrid working model where
employees work from their assigned office on Tuesday, Wednesday,
and Thursday, while Monday and Friday remain remote-flexible.
Fully remote arrangements are still possible by manager exception
for roles hired as remote-only.

Mem0 det-017 [Temporal Reasoning] FAIL — The candidate fails to
provide the specific reimbursement cap of $50/day as stated in
the reference.
 Q: What is the daily meal expense reimbursement cap as of 2024-
01-01? (Answer as of 2024-01-01)
 expected: $50/day
 got     : The facts provided do not include any information
regarding the daily meal expense reimbursement cap as of January
1, 2024. Therefore, I cannot provide an answer to that question.

Mem0 det-018 [Temporal Reasoning] FAIL — The candidate
incorrectly states that the reimbursement cap is unspecified,
while the reference clearly states it is $75/day.
 Q: What is the daily meal expense reimbursement cap as of 2025-
12-01? (Answer as of 2025-12-01)
 expected: $75/day
 got     : The daily meal expense reimbursement cap is not
specified in the provided facts. Therefore, it cannot be
determined as of December 1, 2025.

Mem0 det-019 [Conflict Resolution] FAIL — The candidate omits the
specific daily meal expense reimbursement cap of $75/day stated
in the reference.
 Q: What is the daily meal expense reimbursement cap?
 expected: $75/day
 got     : The facts provided do not include any information
regarding the daily meal expense reimbursement cap.

Mem0 det-020 [Temporal Reasoning] FAIL — The candidate fails to
provide the specific battery life information that is available
in the reference.
 Q: What is the battery life of the Solvane Nav Kit as of 2025-
06-01? (Answer as of 2025-06-01)
 expected: 8 hours standard mode; up to 10 hours in eco mode
(both true, different modes)
 got     : The facts provided do not include any information
regarding the battery life of the Solvane Nav Kit. Therefore, I
cannot provide an answer for that specific detail as of June 1,
2025.

Mem0 det-021 [Conflict Resolution] FAIL — The candidate omits the
specific battery life values provided in the reference answer.
 Q: What is the battery life of the Solvane Nav Kit?
 expected: Both values are current and true simultaneously (8h
standard mode; up to 10h eco mode) — there is no single 'most
recent supersedes all' answer here.
 got     : The battery life of the Solvane Nav Kit is not
mentioned in the provided facts.

Mem0 det-mh-001 [Multi-hop Reasoning] FAIL — The candidate
incorrectly states that Sofia Kappler led the review instead of
Derek Osei.
 Q: Who reviewed the S1 payload issue that led to a leadership
change, and what is the corrected payload rating that came out of
it?
 expected: Derek Osei led the Team Forge review (becoming Team
Forge lead as a result, per the Sept 2024 incident report); the
corrected S1 v2 payload rating is 6 kg (per the Nov 2024 errata).
 got     : The review of the S1 payload issue was conducted by
Team Forge, led by Sofia Kappler. The corrected payload rating
for the Solvane S1 v2 hardware is 6 kg.

Mem0 det-mh-002 [Multi-hop Reasoning] PASS — The candidate
accurately states that Alice Chen is no longer the point of
contact for Team Nova and provides correct details about Bob
Okafor's succession and Alice's new role.
 Q: As of mid-2025, is Alice Chen still the point of contact for
Team Nova? If not, why not and who is?
 expected: No — as of March 2025, Bob Okafor is Team Nova lead.
Alice Chen moved to a Principal Engineer (individual contributor)
role with no direct reports.
 got     : As of mid-2025, Alice Chen is not the point of contact
for Team Nova. Bob Okafor succeeded her as Lead of Team Nova,
effective March 3, 2025, becoming the primary point of contact
for the perception and computer-vision team. Alice Chen moved
into a Principal Engineer role on the same date, focusing on
cross-team technical strategy without direct reports.

Mem0 det-mh-003 [Multi-hop Reasoning] PASS — The candidate
accurately reflects the information from the reference answer
regarding the leadership of Team Forge.
 Q: According to the April 2024 org snapshot, who leads Team
Forge — and is that still accurate?
 expected: The April 2024 snapshot lists Sofia Kappler as Team
Forge lead, which was accurate at that time but is now stale —
Derek Osei became lead in September 2024.
 got     : According to the April 2024 org snapshot, Sofia
Kappler led Team Forge. However, this is no longer accurate as
Derek Osei became the Lead of Team Forge, effective September 16,
2024.

Mem0 det-mh-004 [Multi-hop Reasoning] FAIL — The candidate answer
omits the reasons for Alice's and Sofia's transitions, which are
essential to the reference.
 Q: Alice Chen and Sofia Kappler both stepped down as team leads.
What new roles did each move into, and what prompted each
transition?
 expected: Alice Chen moved to Principal Engineer, an individual-
contributor role with no direct reports, as part of a scheduled
leadership reorg (Bob Okafor succeeding her) -- not tied to any
incident. Sofia Kappler moved to a newly created Quality Systems
Advisor role, specifically prompted by the Team Forge review
following S1 payload field reports. The two transitions had
different causes: routine reorg for Alice, incident-driven review
for Sofia.
 got     : Alice Chen moved into a Principal Engineer role on
March 3, 2025, focusing on cross-team technical strategy for
perception systems across the S1 and future product lines. Sofia
Kappler transitioned to a newly created Quality Systems Advisor
role on September 16, 2024, focusing on hardware validation
across all product lines.

Mem0 det-mh-005 [Multi-hop Reasoning] PASS — The candidate
accurately states that the correction applies only to the June
2024 v2.0 spec sheet and confirms the original 2023 v1.0 spec
sheet remains rated at 5 kg.
 Q: Does the November 2024 S1 payload correction to 6 kg apply to
units running the original 2023 v1.0 hardware?
 expected: No. The errata explicitly corrects only the June 2024
v2.0 spec sheet's 7 kg figure; it does not affect the original
2023 v1.0 spec sheet, which remains rated at 5 kg.
 got     : No, the correction to the 6 kg payload rating applies
solely to the June 2024 v2 spec sheet and does not affect the
original 2023 v1.0 spec sheet, which remains rated at 5 kg.

Mem0 det-mh-006 [Multi-hop Reasoning] FAIL — The candidate
incorrectly states that there is no information about Team Atlas,
while the reference clearly indicates that there has been no
leadership change.
 Q: Team Nova and Team Forge both got new leads at different
points. Did Team Atlas ever get a new lead too?
 expected: No -- Marcus Webb has led Team Atlas continuously
since at least the 2023 handbook, and is still listed as Team
Atlas lead as recently as the October 2024 all-hands notes and
the April 2024 org snapshot. Unlike Nova and Forge, there is no
leadership change recorded for Team Atlas anywhere in the
dataset.
 got     : There is no information provided about Team Atlas or
any changes in its leadership. Therefore, it is unclear if Team
Atlas ever got a new lead.

Mem0 det-mh-007 [Multi-hop Reasoning] FAIL — The candidate
incorrectly states that there was no announcement at the October
2024 all-hands.
 Q: Was the S1 v2 payload correction first made public at the
October 2024 all-hands, or was it announced there before the
official notice went out?
 expected: It was previewed internally at the October 22, 2024
all-hands (Derek walked through the correction ahead of the
public notice), but the official public errata wasn't posted
until November 20, 2024 -- about a month later.
 got     : The S1 v2 payload correction was first made public
through an errata notice posted on November 20, 2024. There is no
indication that it was announced at the October 2024 all-hands
before the official notice went out.

Mem0 det-mh-008 [Multi-hop Reasoning] FAIL — The candidate states
that Bob Okafor is currently the lead, but the reference
indicates that this information is no longer current.
 Q: The employee handbook's original org chart lists Team Nova's
lead as 'TBD.' Who ended up filling that role, and is that still
accurate today?
 expected: Alice Chen was announced as Team Nova's first lead in
January 2024, filling the 'TBD' slot from the 2023 handbook.
That's no longer current, though -- Bob Okafor succeeded her in
March 2025.
 got     : Bob Okafor ended up filling the role of Team Nova's
lead. This is still accurate today, as he succeeded Alice Chen as
Lead of Team Nova effective March 3, 2025.

Mem0 overall: 15/29 (51.7%)
  Temporal Reasoning: 8/14 (57.1%)
  Conflict Resolution: 4/7 (57.1%)
  Multi-hop Reasoning: 3/8 (37.5%)
Supermemory det-001 [Temporal Reasoning] PASS — exact/substring
match
 Q: Who leads Team Nova as of 2024-06-01? (Answer as of 2024-06-
01)
 expected: Alice Chen
 got     : As of 2024-06-01, Alice Chen is the Lead of Team Nova,
effective January 8, 2024. This fact has not been superseded.

Supermemory det-002 [Temporal Reasoning] PASS — exact/substring
match
 Q: Who leads Team Nova as of 2025-06-01? (Answer as of 2025-06-
01)
 expected: Bob Okafor
 got     : Bob Okafor is the Lead of Team Nova, effective March
3, 2025.

Supermemory det-003 [Conflict Resolution] FAIL — The candidate
states a different leader for Team Nova than the reference.
 Q: Who leads Team Nova?
 expected: Bob Okafor
 got     : Alice Chen is the Lead of Team Nova, effective January
8, 2024.

Supermemory det-004 [Temporal Reasoning] PASS — exact/substring
match
 Q: Who leads Team Forge as of 2024-03-01? (Answer as of 2024-03-
01)
 expected: Sofia Kappler
 got     : Sofia Kappler leads Team Forge as of 2024-03-01. This
information is still accurate as of that date and has not yet
been superseded.

Supermemory det-005 [Temporal Reasoning] PASS — exact/substring
match
 Q: Who leads Team Forge as of 2025-01-01? (Answer as of 2025-01-
01)
 expected: Derek Osei
 got     : Derek Osei is Lead of Team Forge effective September
16, 2024.

Supermemory det-006 [Conflict Resolution] PASS — exact/substring
match
 Q: Who leads Team Forge?
 expected: Derek Osei
 got     : Derek Osei is now Lead of Team Forge, effective
September 16, 2024.

Supermemory det-007 [Temporal Reasoning] FAIL — The candidate
states a different maximum payload (7 kg) than the reference (5
kg).
 Q: What is the maximum payload of the Solvane S1 as of 2023-06-
01? (Answer as of 2023-06-01)
 expected: 5 kg
 got     : The maximum payload of the Solvane S1 v2 as of 2023-
06-01 is 7 kilograms. This fact is still accurate as of that
date.

Supermemory det-008 [Temporal Reasoning] FAIL — The candidate
states a different maximum payload than the reference answer.
 Q: What is the maximum payload of the Solvane S1 as of 2024-08-
01? (Answer as of 2024-08-01)
 expected: 7 kg
 got     : The maximum payload of the Solvane S1 as of 2024-08-01
is 6 kg under real-world conditions. This is based on the updated
specifications that took effect before this date.

Supermemory det-009 [Temporal Reasoning] PASS — exact/substring
match
 Q: What is the maximum payload of the Solvane S1 as of 2025-06-
01? (Answer as of 2025-06-01)
 expected: 6 kg
 got     : The maximum payload of the Solvane S1 as of 2025-06-01
is 6 kg under real-world conditions. This is based on the updated
specifications effective from 2024-11-20, which superseded the
previous maximum payload of 7 kg.

Supermemory det-010 [Conflict Resolution] PASS — exact/substring
match
 Q: What is the maximum payload of the Solvane S1?
 expected: 6 kg
 got     : The maximum payload of the Solvane S1 v2 is 6 kg under
real-world conditions.

Supermemory det-011 [Temporal Reasoning] PASS — The candidate's
core factual claim about the pricing matches the reference
answer.
 Q: What is the pricing for Solvane Vision Suite as of 2024-01-
01? (Answer as of 2024-01-01)
 expected: $499/month flat, individual license
 got     : As of January 1, 2024, the pricing for Solvane Vision
Suite is $499 per month, flat rate, individual license. This was
effective from May 1, 2023, and has not yet been superseded by
the per-seat pricing that takes effect on January 15, 2025.

Supermemory det-012 [Temporal Reasoning] PASS — The candidate's
core factual claim about the pricing matches the reference
answer.
 Q: What is the pricing for Solvane Vision Suite as of 2025-06-
01? (Answer as of 2025-06-01)
 expected: $150/seat/month, per-seat license
 got     : As of 2025-06-01, the pricing for Solvane Vision Suite
is $150 per seat per month, following the transition to per-seat
pricing effective January 15, 2025. The previous flat-rate
individual tier of $499 per month has been discontinued.

Supermemory det-013 [Conflict Resolution] PASS — The core factual
claim about the pricing matches the reference answer.
 Q: What is the pricing for Solvane Vision Suite?
 expected: $150/seat/month, per-seat license (individual tier
discontinued)
 got     : The pricing for Solvane Vision Suite is $150 per seat
per month, effective January 15, 2025. The previous flat-rate
individual tier of $499 per month has been discontinued.

Supermemory det-014 [Temporal Reasoning] FAIL — The candidate
states that employees are expected to work in-office on specific
days, contradicting the reference's claim of a fully remote
policy.
 Q: What is Solvane's remote work policy as of 2024-01-01?
(Answer as of 2024-01-01)
 expected: Fully remote, no in-office requirement
 got     : As of January 1, 2024, Solvane Robotics' remote work
policy is that employees are expected to work from an assigned
office on Tuesday, Wednesday, and Thursday, with remote work on
Monday and Friday being flexible. Fully remote arrangements are
available by manager exception for remote-only roles.

Supermemory det-015 [Temporal Reasoning] FAIL — The candidate's
answer includes additional remote work days (Monday and Friday)
that are not part of the reference answer.
 Q: What is Solvane's remote work policy as of 2025-06-01?
(Answer as of 2025-06-01)
 expected: Hybrid: 3 days/week in office (Tue/Wed/Thu)
 got     : As of June 1, 2025, Solvane Robotics' remote work
policy is that employees are expected to work from an assigned
office on Tuesday, Wednesday, and Thursday, and remotely on
Monday and Friday. Fully remote arrangements are still available
by manager exception for remote-only roles.

Supermemory det-016 [Conflict Resolution] PASS — The candidate
accurately describes the in-office days as Tuesday, Wednesday,
and Thursday, aligning with the reference answer.
 Q: What is Solvane's remote work policy?
 expected: Hybrid: 3 days/week in office (Tue/Wed/Thu)
 got     : Solvane Robotics is moving to a hybrid working model;
employees are expected to work from an assigned office on
Tuesday, Wednesday, and Thursday, and remotely on Monday and
Friday, with fully remote arrangements still available by manager
exception, effective February 1, 2025.

Supermemory det-017 [Temporal Reasoning] PASS — The candidate
accurately states the daily meal expense reimbursement cap as
$50/day.
 Q: What is the daily meal expense reimbursement cap as of 2024-
01-01? (Answer as of 2024-01-01)
 expected: $50/day
 got     : The daily meal expense reimbursement cap as of 2024-
01-01 is $50 per day, with receipts required for any single
expense over $25. This policy is still in effect and has not yet
been superseded by the increase to $75 per day, which takes
effect on August 1, 2025.

Supermemory det-018 [Temporal Reasoning] PASS — The candidate's
core factual claim matches the reference answer regarding the
daily meal expense reimbursement cap.
 Q: What is the daily meal expense reimbursement cap as of 2025-
12-01? (Answer as of 2025-12-01)
 expected: $75/day
 got     : The daily meal expense reimbursement cap as of 2025-
12-01 is $75 per day. This is effective from August 1, 2025, and
reflects the updated policy.

Supermemory det-019 [Conflict Resolution] PASS — The candidate's
answer matches the reference answer in terms of the daily meal
expense reimbursement cap.
 Q: What is the daily meal expense reimbursement cap?
 expected: $75/day
 got     : The daily meal expense reimbursement cap is $75 per
day.

Supermemory det-020 [Temporal Reasoning] FAIL — The candidate
answer omits the standard mode battery life of 8 hours.
 Q: What is the battery life of the Solvane Nav Kit as of 2025-
06-01? (Answer as of 2025-06-01)
 expected: 8 hours standard mode; up to 10 hours in eco mode
(both true, different modes)
 got     : The battery life of the Solvane Nav Kit in eco mode is
up to 10 hours continuous operation as of 2025-06-01. This fact
has not been superseded.

Supermemory det-021 [Conflict Resolution] PASS — The candidate
accurately states the battery life for both standard and eco
modes as per the reference.
 Q: What is the battery life of the Solvane Nav Kit?
 expected: Both values are current and true simultaneously (8h
standard mode; up to 10h eco mode) — there is no single 'most
recent supersedes all' answer here.
 got     : The battery life of the Solvane Nav Kit in eco mode is
up to 10 hours of continuous operation. In standard operating
mode, the Solvane Nav Kit v1.0 has a battery life of 8 hours.

Supermemory det-mh-001 [Multi-hop Reasoning] FAIL — The candidate
does not specify that Derek Osei led the review, which is a key
detail from the reference answer.
 Q: Who reviewed the S1 payload issue that led to a leadership
change, and what is the corrected payload rating that came out of
it?
 expected: Derek Osei led the Team Forge review (becoming Team
Forge lead as a result, per the Sept 2024 incident report); the
corrected S1 v2 payload rating is 6 kg (per the Nov 2024 errata).
 got     : The S1 payload issue was reviewed by Team Forge, and
the corrected maximum payload rating that came out of it is 6 kg
under real-world deployment conditions.

Supermemory det-mh-002 [Multi-hop Reasoning] PASS — The candidate
accurately states that Alice Chen is no longer the point of
contact and provides correct details about her transition and Bob
Okafor's new role.
 Q: As of mid-2025, is Alice Chen still the point of contact for
Team Nova? If not, why not and who is?
 expected: No — as of March 2025, Bob Okafor is Team Nova lead.
Alice Chen moved to a Principal Engineer (individual contributor)
role with no direct reports.
 got     : As of mid-2025, Alice Chen is not the point of contact
for Team Nova. She transitioned to Principal Engineer on March 3,
2025, and Bob Okafor became the Lead of Team Nova effective March
3, 2025.

Supermemory det-mh-003 [Multi-hop Reasoning] FAIL — The candidate
fails to acknowledge the specific information about Sofia Kappler
being the lead and the subsequent change to Derek Osei.
 Q: According to the April 2024 org snapshot, who leads Team
Forge — and is that still accurate?
 expected: The April 2024 snapshot lists Sofia Kappler as Team
Forge lead, which was accurate at that time but is now stale —
Derek Osei became lead in September 2024.
 got     : No facts are available to provide an answer regarding
who leads Team Forge or whether that information is still
accurate.

Supermemory det-mh-004 [Multi-hop Reasoning] FAIL — The candidate
omits the specific reasons for the transitions, stating they were
prompted by new roles rather than the scheduled leadership reorg
for Alice and the incident-driven review for Sofia.
 Q: Alice Chen and Sofia Kappler both stepped down as team leads.
What new roles did each move into, and what prompted each
transition?
 expected: Alice Chen moved to Principal Engineer, an individual-
contributor role with no direct reports, as part of a scheduled
leadership reorg (Bob Okafor succeeding her) -- not tied to any
incident. Sofia Kappler moved to a newly created Quality Systems
Advisor role, specifically prompted by the Team Forge review
following S1 payload field reports. The two transitions had
different causes: routine reorg for Alice, incident-driven review
for Sofia.
 got     : Alice Chen transitioned to Principal Engineer,
focusing on cross-team technical strategy for perception systems
across the S1 and future product lines, effective March 3, 2025.
Sofia Kappler transitioned to Quality Systems Advisor, focusing
on hardware validation across all product lines, effective
September 16, 2024. The transitions were prompted by their
respective new roles and responsibilities within the
organization.

Supermemory det-mh-005 [Multi-hop Reasoning] PASS — The candidate
accurately states that the correction does not apply to the
original 2023 v1.0 hardware and confirms its maximum payload of 5
kg.
 Q: Does the November 2024 S1 payload correction to 6 kg apply to
units running the original 2023 v1.0 hardware?
 expected: No. The errata explicitly corrects only the June 2024
v2.0 spec sheet's 7 kg figure; it does not affect the original
2023 v1.0 spec sheet, which remains rated at 5 kg.
 got     : No, the November 2024 S1 payload correction to 6 kg
does not apply to units running the original 2023 v1.0 hardware,
which remains at a maximum payload of 5 kg.

Supermemory det-mh-006 [Multi-hop Reasoning] PASS — The candidate
accurately states that Team Atlas has not gotten a new lead and
confirms Marcus Webb is still the lead.
 Q: Team Nova and Team Forge both got new leads at different
points. Did Team Atlas ever get a new lead too?
 expected: No -- Marcus Webb has led Team Atlas continuously
since at least the 2023 handbook, and is still listed as Team
Atlas lead as recently as the October 2024 all-hands notes and
the April 2024 org snapshot. Unlike Nova and Forge, there is no
leadership change recorded for Team Atlas anywhere in the
dataset.
 got     : No, Team Atlas has not gotten a new lead; Marcus Webb
is still the lead.

Supermemory det-mh-007 [Multi-hop Reasoning] PASS — The candidate
accurately states that the correction was announced at the
October 2024 all-hands before the official notice.
 Q: Was the S1 v2 payload correction first made public at the
October 2024 all-hands, or was it announced there before the
official notice went out?
 expected: It was previewed internally at the October 22, 2024
all-hands (Derek walked through the correction ahead of the
public notice), but the official public errata wasn't posted
until November 20, 2024 -- about a month later.
 got     : The S1 v2 payload correction was announced at the
October 2024 all-hands before the official notice went out.

Supermemory det-mh-008 [Multi-hop Reasoning] FAIL — The candidate
fails to provide the specific information about Team Nova's lead
and its accuracy, which contradicts the reference.
 Q: The employee handbook's original org chart lists Team Nova's
lead as 'TBD.' Who ended up filling that role, and is that still
accurate today?
 expected: Alice Chen was announced as Team Nova's first lead in
January 2024, filling the 'TBD' slot from the 2023 handbook.
That's no longer current, though -- Bob Okafor succeeded her in
March 2025.
 got     : No facts are available to provide an answer regarding
Team Nova's lead or the accuracy of that information.

Supermemory overall: 19/29 (65.5%)
  Temporal Reasoning: 9/14 (64.3%)
  Conflict Resolution: 6/7 (85.7%)
  Multi-hop Reasoning: 4/8 (50.0%)
Cognee det-001 [Temporal Reasoning] PASS — exact/substring match
 Q: Who leads Team Nova as of 2024-06-01? (Answer as of 2024-06-
01)
 expected: Alice Chen
 got     : As of June 1, 2024, Alice Chen leads Team Nova. This
was confirmed in the announcement made on January 8, 2024.

Cognee det-002 [Temporal Reasoning] PASS — exact/substring match
 Q: Who leads Team Nova as of 2025-06-01? (Answer as of 2025-06-
01)
 expected: Bob Okafor
 got     : As of June 1, 2025, Bob Okafor leads Team Nova. This
change took effect on March 3, 2025, when he succeeded Alice
Chen, who transitioned to a Principal Engineer role.

Cognee det-003 [Conflict Resolution] PASS — exact/substring match
 Q: Who leads Team Nova?
 expected: Bob Okafor
 got     : Bob Okafor leads Team Nova.

Cognee det-004 [Temporal Reasoning] PASS — exact/substring match
 Q: Who leads Team Forge as of 2024-03-01? (Answer as of 2024-03-
01)
 expected: Sofia Kappler
 got     : As of March 1, 2024, Team Forge is led by Sofia
Kappler. This information is still accurate as of that date, as
Derek Osei's transition to Lead of Team Forge occurred later,
effective September 16, 2024.

Cognee det-005 [Temporal Reasoning] PASS — exact/substring match
 Q: Who leads Team Forge as of 2025-01-01? (Answer as of 2025-01-
01)
 expected: Derek Osei
 got     : Derek Osei leads Team Forge as of January 1, 2025.

Cognee det-006 [Conflict Resolution] PASS — exact/substring match
 Q: Who leads Team Forge?
 expected: Derek Osei
 got     : Derek Osei leads Team Forge.

Cognee det-007 [Temporal Reasoning] PASS — exact/substring match
 Q: What is the maximum payload of the Solvane S1 as of 2023-06-
01? (Answer as of 2023-06-01)
 expected: 5 kg
 got     : The maximum payload of the Solvane S1 as of June 1,
2023, is 5 kg, according to the original specification sheet for
version 1.0 published in February 2023. This information is still
accurate as of that date.

Cognee det-008 [Temporal Reasoning] PASS — exact/substring match
 Q: What is the maximum payload of the Solvane S1 as of 2024-08-
01? (Answer as of 2024-08-01)
 expected: 7 kg
 got     : The maximum payload of the Solvane S1 as of 2024-08-01
is 7 kg, according to the Solvane S1 v2.0 spec sheet published on
June 3, 2024. This information has not yet been superseded by any
subsequent updates as of that date.

Cognee det-009 [Temporal Reasoning] PASS — exact/substring match
 Q: What is the maximum payload of the Solvane S1 as of 2025-06-
01? (Answer as of 2025-06-01)
 expected: 6 kg
 got     : The maximum payload of the Solvane S1 as of June 1,
2025, is 6 kg. This is based on the errata notice issued on
November 20, 2024, which corrected the previously published
maximum payload of 7 kg from the Solvane S1 v2.0 spec sheet
released on June 3, 2024.

Cognee det-010 [Conflict Resolution] PASS — exact/substring match
 Q: What is the maximum payload of the Solvane S1?
 expected: 6 kg
 got     : The maximum payload of the Solvane S1 is 6 kg under
real-world conditions, as updated by the errata notice on
November 20, 2024. The previous maximum payload listed was 7 kg
based on controlled lab conditions, but this has since been
corrected.

Cognee det-011 [Temporal Reasoning] PASS — The candidate
accurately states the core pricing of $499/month for an
individual license.
 Q: What is the pricing for Solvane Vision Suite as of 2024-01-
01? (Answer as of 2024-01-01)
 expected: $499/month flat, individual license
 got     : As of January 1, 2024, the pricing for the Solvane
Vision Suite is as follows: the Standard tier costs $499 per
month, flat rate, for an individual license covering a single
deployment instance with standard support. Enterprise pricing is
available on request for multi-site deployments. This pricing has
not changed since it was outlined on May 1, 2023.

Cognee det-012 [Temporal Reasoning] FAIL — The candidate states a
different price than the reference answer.
 Q: What is the pricing for Solvane Vision Suite as of 2025-06-
01? (Answer as of 2025-06-01)
 expected: $150/seat/month, per-seat license
 got     : As of June 1, 2025, the pricing for the Solvane Vision
Suite is still $499 per month for the Standard tier, which covers
a single deployment instance with standard support. Enterprise
pricing remains available on request for multi-site deployments.
This pricing has not changed since it was outlined on May 1,
2023.
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Cognee det-013 [Conflict Resolution] FAIL — The candidate states
a different pricing structure than the reference answer.
 Q: What is the pricing for Solvane Vision Suite?
 expected: $150/seat/month, per-seat license (individual tier
discontinued)
 got     : The pricing for Solvane Vision Suite is as follows:
the Standard tier costs $499 per month, flat rate, for an
individual license covering a single deployment instance with
standard support. Enterprise pricing is available on request for
multi-site deployments.

Cognee det-014 [Temporal Reasoning] PASS — The candidate's answer
accurately reflects the reference's core claim about Solvane's
remote work policy.
 Q: What is Solvane's remote work policy as of 2024-01-01?
(Answer as of 2024-01-01)
 expected: Fully remote, no in-office requirement
 got     : As of January 1, 2024, Solvane Robotics is a fully
remote company with no mandatory in-office days; optional in-
person meetings are at team discretion. This policy is still
accurate and has not been superseded by any updates as of that
date.

Cognee det-015 [Temporal Reasoning] PASS — The candidate
accurately describes the hybrid model of 3 days in the
office (Tue/Wed/Thu) while adding extra details about remote
flexibility and exceptions.
 Q: What is Solvane's remote work policy as of 2025-06-01?
(Answer as of 2025-06-01)
 expected: Hybrid: 3 days/week in office (Tue/Wed/Thu)
 got     : As of June 1, 2025, Solvane Robotics has adopted a
hybrid working model. Employees are expected to work on-site at
an assigned Solvane office three days per week (Tuesday,
Wednesday, and Thursday), while Monday and Friday remain remote-
flexible for all teams. Fully remote arrangements are still
available by manager exception for roles hired as remote-only.

Cognee det-016 [Conflict Resolution] FAIL — The candidate states
that Solvane is currently fully remote, which contradicts the
reference's claim of a hybrid model with specific in-office days.
 Q: What is Solvane's remote work policy?
 expected: Hybrid: 3 days/week in office (Tue/Wed/Thu)
 got     : Solvane Robotics is a fully remote company with no
mandatory in-office days; optional in-person meetings are at team
discretion. However, as of February 1, 2025, the company is
moving to a hybrid working model where employees are expected to
work on-site three days per week, with remote work available on
Monday and Friday under a remote-flexible policy. Fully remote
arrangements remain available only by manager exception for roles
hired as remote-only.

Cognee det-017 [Temporal Reasoning] PASS — The candidate
accurately states the daily meal expense reimbursement cap as
$50/day.
 Q: What is the daily meal expense reimbursement cap as of 2024-
01-01? (Answer as of 2024-01-01)
 expected: $50/day
 got     : As of January 1, 2024, the daily meal expense
reimbursement cap for business travel is $50 per day, as stated
in the Employee Handbook effective January 1, 2023. This fact has
not yet been superseded by the memo from August 1, 2025, which
raises the cap to $75 per day.

Cognee det-018 [Temporal Reasoning] PASS — The candidate's core
factual claim matches the reference answer regarding the daily
meal expense reimbursement cap.
 Q: What is the daily meal expense reimbursement cap as of 2025-
12-01? (Answer as of 2025-12-01)
 expected: $75/day
 got     : The daily meal expense reimbursement cap as of
December 1, 2025, is $75 per day. This reflects an increase from
the previous cap of $50 per day, which was in effect prior to the
update announced in the finance memo on August 1, 2025.

Cognee det-019 [Conflict Resolution] PASS — The candidate's
answer matches the reference answer regarding the daily meal
expense reimbursement cap.
 Q: What is the daily meal expense reimbursement cap?
 expected: $75/day
 got     : The daily meal expense reimbursement cap for business
travel is $75 per day.

Cognee det-020 [Temporal Reasoning] FAIL — The candidate omits
the eco mode battery life of up to 10 hours.
 Q: What is the battery life of the Solvane Nav Kit as of 2025-
06-01? (Answer as of 2025-06-01)
 expected: 8 hours standard mode; up to 10 hours in eco mode
(both true, different modes)
 got     : The battery life of the Solvane Nav Kit is 8 hours of
continuous operation in standard operating mode. This information
remains accurate as of June 1, 2025.

Cognee det-021 [Conflict Resolution] FAIL — The candidate answer
omits the eco mode battery life of up to 10 hours.
 Q: What is the battery life of the Solvane Nav Kit?
 expected: Both values are current and true simultaneously (8h
standard mode; up to 10h eco mode) — there is no single 'most
recent supersedes all' answer here.
 got     : The battery life of the Solvane Nav Kit is 8 hours of
continuous operation in standard operating mode.

Cognee det-mh-001 [Multi-hop Reasoning] FAIL — The candidate
answer does not specify that Derek Osei led the Team Forge
review, which is a key detail from the reference.
 Q: Who reviewed the S1 payload issue that led to a leadership
change, and what is the corrected payload rating that came out of
it?
 expected: Derek Osei led the Team Forge review (becoming Team
Forge lead as a result, per the Sept 2024 incident report); the
corrected S1 v2 payload rating is 6 kg (per the Nov 2024 errata).
 got     : The S1 payload issue was reviewed by Team Forge, and
the corrected payload rating for S1 v2 hardware is 6 kg.

Cognee det-mh-002 [Multi-hop Reasoning] PASS — The candidate
accurately states that Alice Chen is no longer the point of
contact and provides correct details about Bob Okafor's
succession and Alice's new role.
 Q: As of mid-2025, is Alice Chen still the point of contact for
Team Nova? If not, why not and who is?
 expected: No — as of March 2025, Bob Okafor is Team Nova lead.
Alice Chen moved to a Principal Engineer (individual contributor)
role with no direct reports.
 got     : As of mid-2025, Alice Chen is not the point of contact
for Team Nova. Bob Okafor succeeded her as the Lead of Team Nova
effective March 3, 2025, and he is now the point of contact for
the team. Alice Chen transitioned to a Principal Engineer role,
focusing on cross-team technical strategy for perception systems.

Cognee det-mh-003 [Multi-hop Reasoning] PASS — The candidate
accurately reflects the information from the reference regarding
the leadership of Team Forge and the timeline of changes.
 Q: According to the April 2024 org snapshot, who leads Team
Forge — and is that still accurate?
 expected: The April 2024 snapshot lists Sofia Kappler as Team
Forge lead, which was accurate at that time but is now stale —
Derek Osei became lead in September 2024.
 got     : According to the April 2024 org snapshot, Sofia
Kappler led Team Forge. However, this is no longer accurate; as
of September 16, 2024, Derek Osei became the Lead of Team Forge.

Cognee det-mh-004 [Multi-hop Reasoning] FAIL — The candidate
incorrectly states that Alice's transition was tied to Bob Okafor
succeeding her, which contradicts the reference that it was part
of a scheduled reorg not tied to any incident.
 Q: Alice Chen and Sofia Kappler both stepped down as team leads.
What new roles did each move into, and what prompted each
transition?
 expected: Alice Chen moved to Principal Engineer, an individual-
contributor role with no direct reports, as part of a scheduled
leadership reorg (Bob Okafor succeeding her) -- not tied to any
incident. Sofia Kappler moved to a newly created Quality Systems
Advisor role, specifically prompted by the Team Forge review
following S1 payload field reports. The two transitions had
different causes: routine reorg for Alice, incident-driven review
for Sofia.
 got     : Alice Chen transitioned to a Principal Engineer role,
focusing on cross-team technical strategy for perception systems.
This change was prompted by Bob Okafor succeeding her as the Lead
of Team Nova, effective March 3, 2025.

Sofia Kappler moved into a Quality Systems Advisor role, where
she focuses on hardware validation processes across all product
lines. Her transition was prompted by the leadership change in
Team Forge, where Derek Osei became the new Lead of Team Forge.

Cognee det-mh-005 [Multi-hop Reasoning] PASS — The candidate
accurately states that the correction does not apply to the
original 2023 v1.0 hardware and confirms its rating of 5 kg.
 Q: Does the November 2024 S1 payload correction to 6 kg apply to
units running the original 2023 v1.0 hardware?
 expected: No. The errata explicitly corrects only the June 2024
v2.0 spec sheet's 7 kg figure; it does not affect the original
2023 v1.0 spec sheet, which remains rated at 5 kg.
 got     : No, the November 2024 S1 payload correction to 6 kg
does not apply to units running the original 2023 v1.0 hardware,
which remains rated at 5 kg.

Cognee det-mh-006 [Multi-hop Reasoning] PASS — The candidate
accurately states that Team Atlas has not had a new lead and is
still led by Marcus Webb.
 Q: Team Nova and Team Forge both got new leads at different
points. Did Team Atlas ever get a new lead too?
 expected: No -- Marcus Webb has led Team Atlas continuously
since at least the 2023 handbook, and is still listed as Team
Atlas lead as recently as the October 2024 all-hands notes and
the April 2024 org snapshot. Unlike Nova and Forge, there is no
leadership change recorded for Team Atlas anywhere in the
dataset.
 got     : No, Team Atlas has not had a new lead; it is still led
by Marcus Webb.

Cognee det-mh-007 [Multi-hop Reasoning] PASS — The candidate
accurately states that the correction was previewed at the
October 2024 all-hands before the official notice.
 Q: Was the S1 v2 payload correction first made public at the
October 2024 all-hands, or was it announced there before the
official notice went out?
 expected: It was previewed internally at the October 22, 2024
all-hands (Derek walked through the correction ahead of the
public notice), but the official public errata wasn't posted
until November 20, 2024 -- about a month later.
 got     : The S1 v2 payload correction was previewed at the
October 2024 all-hands meeting before the official notice went
out.

Cognee det-mh-008 [Multi-hop Reasoning] FAIL — The candidate
states that Bob Okafor is currently the lead, but the reference
indicates that this information is no longer current.
 Q: The employee handbook's original org chart lists Team Nova's
lead as 'TBD.' Who ended up filling that role, and is that still
accurate today?
 expected: Alice Chen was announced as Team Nova's first lead in
January 2024, filling the 'TBD' slot from the 2023 handbook.
That's no longer current, though -- Bob Okafor succeeded her in
March 2025.
 got     : Bob Okafor ended up filling the role of Lead of Team
Nova. This is still accurate today, as he succeeded Alice Chen in
that position effective March 3, 2025.

Cognee overall: 21/29 (72.4%)
  Temporal Reasoning: 12/14 (85.7%)
  Conflict Resolution: 4/7 (57.1%)
  Multi-hop Reasoning: 5/8 (62.5%)
Zep persona-01 PASS — exact/substring match
 Q: What day and time is Priya's dentist appointment?
 expected: Thursday at 2pm
 got     : Priya's dentist appointment is scheduled for Thursday
at 2pm.

Zep persona-02 PASS — The candidate's answer matches the
reference answer regarding the date of Sam's nephew's birthday.
 Q: When is Sam's nephew's birthday?
 expected: March 15
 got     : Sam's nephew's birthday is on March 15th.

Zep persona-03 PASS — exact/substring match
 Q: What is the name of Jordan's cat?
 expected: Mochi
 got     : The name of Jordan's cat is Mochi.

Zep persona-04 FAIL — The candidate states that the address is
unknown, which contradicts the reference answer.
 Q: What is Noor's new address?
 expected: 42 Ridge Lane
 got     : Noor's new address is unknown.

Zep persona-05 PASS — exact/substring match
 Q: What car did Devon get from his sister?
 expected: a blue Honda Civic
 got     : Devon received a blue Honda Civic from their sister.

Zep persona-06 FAIL — The candidate states Biscuit's age is
unknown, contradicting the reference that specifies Biscuit was 3
years old.
 Q: How old was Biscuit as of the first session (January 2026)?
 expected: 3
 got     : Biscuit's age as of the first session in January 2026
is unknown.

Zep persona-06 FAIL — The candidate answer contradicts the
reference by stating the plant has not been identified, while the
reference specifies it is the pothos.
 Q: Which specific plant did Biscuit knock off the shelf?
 expected: the pothos
 got     : The specific plant that Biscuit knocked off the shelf
has not been identified.

Zep persona-07 PASS — exact/substring match
 Q: Which day is Morgan out of office for a conference?
 expected: next Wednesday
 got     : Morgan Persona is out of office for a conference next
Wednesday.

Zep persona-08 PASS — exact/substring match
 Q: What is Elena's phone number?
 expected: 555-0142
 got     : Elena's phone number is 555-0142.

Zep persona-09 FAIL — The candidate states a different highway
(I-95) than the reference (I-35).
 Q: What highway does Theo mention having a bad commute on?
 expected: I-35
 got     : Theo mentions having a bad commute on I-95.

Zep persona-10 FAIL — The candidate only mentions two meeting
dates instead of three as stated in the reference.
 Q: How many different meeting dates have I heard, and from whom?
 expected: Three: Ben said Thursday, Priya said Tuesday, and
Marcus said Wednesday.
 got     : Aisha Persona has heard about two different meeting
dates regarding the client meeting:

1. Next Thursday (confirmed by Ben).
2. Tuesday (as mentioned by Priya).

Ben confirmed the meeting for next Thursday, while Priya stated
it is staying on the original date, Tuesday.

Zep persona-11 PASS — exact/substring match
 Q: What is Ravi's badge number?
 expected: B-4471
 got     : Ravi's badge number is B-4471.

Zep: 7/12 (58.3%)
Mem0 persona-01 PASS — The core factual claim about the day and
time of the appointment matches the reference.
 Q: What day and time is Priya's dentist appointment?
 expected: Thursday at 2pm
 got     : Priya's dentist appointment is on Thursday, July 30,
2026, at 2 p.m.

Mem0 persona-02 PASS — exact/substring match
 Q: When is Sam's nephew's birthday?
 expected: March 15
 got     : Sam's nephew's birthday is on March 15.

Mem0 persona-03 PASS — exact/substring match
 Q: What is the name of Jordan's cat?
 expected: Mochi
 got     : Jordan's cat is named Mochi.

Mem0 persona-04 PASS — exact/substring match
 Q: What is Noor's new address?
 expected: 42 Ridge Lane
 got     : Noor's new mailing address is 42 Ridge Lane.

Mem0 persona-05 PASS — exact/substring match
 Q: What car did Devon get from his sister?
 expected: a blue Honda Civic
 got     : Devon acquired a blue Honda Civic from his sister.

Mem0 persona-06 PASS — exact/substring match
 Q: How old was Biscuit as of the first session (January 2026)?
 expected: 3
 got     : As of January 2026, Biscuit was 2 years old, turning 3
on July 24, 2026.

Mem0 persona-06 PASS — The candidate's answer matches the
reference by identifying the specific plant as a pothos.
 Q: Which specific plant did Biscuit knock off the shelf?
 expected: the pothos
 got     : Biscuit knocked a pothos plant off the shelf.

Mem0 persona-07 FAIL — The candidate specifies a date that does
not match "next Wednesday" as referenced.
 Q: Which day is Morgan out of office for a conference?
 expected: next Wednesday
 got     : Morgan will be out of the office on Wednesday, July
29, 2026, to attend a conference.

Mem0 persona-08 PASS — exact/substring match
 Q: What is Elena's phone number?
 expected: 555-0142
 got     : Elena's phone number is 555-0142.

Mem0 persona-09 PASS — exact/substring match
 Q: What highway does Theo mention having a bad commute on?
 expected: I-35
 got     : Theo mentions having a bad commute on I-35.

Mem0 persona-10 FAIL — The candidate provides specific dates that
do not match the reference answer's days of the week.
 Q: How many different meeting dates have I heard, and from whom?
 expected: Three: Ben said Thursday, Priya said Tuesday, and
Marcus said Wednesday.
 got     : You have heard of three different meeting dates from
the following sources:

1. Wednesday, July 29, 2026 — from Marcus in operations.
2. Thursday, July 30, 2026 — from coworker Ben.
3. Tuesday, July 28, 2026 — from Priya on July 24, 2026, and
confirmed again on July 29, 2026.

Mem0 persona-11 PASS — exact/substring match
 Q: What is Ravi's badge number?
 expected: B-4471
 got     : The user's badge number for building access is B-4471.

Mem0: 10/12 (83.3%)
Supermemory persona-01 PASS — exact/substring match
 Q: What day and time is Priya's dentist appointment?
 expected: Thursday at 2pm
 got     : Priya's dentist appointment is scheduled for Thursday
at 2pm.

Supermemory persona-02 PASS — exact/substring match
 Q: When is Sam's nephew's birthday?
 expected: March 15
 got     : Sam's nephew's birthday is March 15.

Supermemory persona-03 PASS — exact/substring match
 Q: What is the name of Jordan's cat?
 expected: Mochi
 got     : The name of Jordan's cat is Mochi.

Supermemory persona-04 PASS — exact/substring match
 Q: What is Noor's new address?
 expected: 42 Ridge Lane
 got     : Noor's new address is 42 Ridge Lane.

Supermemory persona-05 PASS — exact/substring match
 Q: What car did Devon get from his sister?
 expected: a blue Honda Civic
 got     : Devon got a blue Honda Civic from his sister.

Supermemory persona-06 FAIL — The candidate fails to provide the
specific age of Biscuit, which is stated as 3 in the reference.
 Q: How old was Biscuit as of the first session (January 2026)?
 expected: 3
 got     : No information is available regarding Biscuit's age as
of January 2026.

Supermemory persona-06 PASS — The candidate's answer matches the
reference by identifying the specific plant as a pothos.
 Q: Which specific plant did Biscuit knock off the shelf?
 expected: the pothos
 got     : Biscuit knocked a pothos plant off the shelf.

Supermemory persona-07 FAIL — The candidate specifies a date that
does not correspond to "next Wednesday."
 Q: Which day is Morgan out of office for a conference?
 expected: next Wednesday
 got     : Morgan is out of office on 2026-07-30 for a
conference.

Supermemory persona-08 PASS — exact/substring match
 Q: What is Elena's phone number?
 expected: 555-0142
 got     : Elena's phone number is 555-0142.

Supermemory persona-09 PASS — exact/substring match
 Q: What highway does Theo mention having a bad commute on?
 expected: I-35
 got     : Theo mentions having a bad commute on I-35.

Supermemory persona-10 FAIL — The candidate's answer includes
specific dates and additional context that alters the core claim
about the meeting dates and sources.
 Q: How many different meeting dates have I heard, and from whom?
 expected: Three: Ben said Thursday, Priya said Tuesday, and
Marcus said Wednesday.
 got     : You have heard of three different meeting dates from
the following sources:

1. Ben confirmed the client meeting is scheduled for next
Thursday (2026-07-28).
2. Marcus in ops indicated the client meeting might be on
Wednesday (2026-07-27), though uncertain.
3. Priya said the client meeting remains scheduled for the
original date, Tuesday (2026-07-26).

The most recent and confirmed date is Tuesday, July 26, 2026, as
updated by the assistant.

Supermemory persona-11 PASS — exact/substring match
 Q: What is Ravi's badge number?
 expected: B-4471
 got     : Ravi's badge number is B-4471.

Supermemory: 9/12 (75.0%)
Cognee persona-01 PASS — The candidate's answer matches the
reference answer exactly.
 Q: What day and time is Priya's dentist appointment?
 expected: Thursday at 2pm
 got     : Priya's dentist appointment is on Thursday at 2 p.m.

Cognee persona-02 PASS — exact/substring match
 Q: When is Sam's nephew's birthday?
 expected: March 15
 got     : Sam's nephew's birthday is on March 15, 2026.

Cognee persona-03 PASS — exact/substring match
 Q: What is the name of Jordan's cat?
 expected: Mochi
 got     : The name of Jordan's cat is Mochi.

Cognee persona-04 PASS — exact/substring match
 Q: What is Noor's new address?
 expected: 42 Ridge Lane
 got     : Noor's new address is 42 Ridge Lane.

Cognee persona-05 FAIL — The candidate answer contradicts the
reference by stating there is no information when there is a
specific answer.
 Q: What car did Devon get from his sister?
 expected: a blue Honda Civic
 got     : There is no information available about Devon
receiving a car from his sister.

Cognee persona-06 PASS — The candidate's answer correctly states
Biscuit was three years old as of January 2026.
 Q: How old was Biscuit as of the first session (January 2026)?
 expected: 3
 got     : Biscuit was three years old as of the first session on
January 8, 2026.

Cognee persona-06 FAIL — The candidate omits the specific plant
mentioned in the reference answer.
 Q: Which specific plant did Biscuit knock off the shelf?
 expected: the pothos
 got     : The specific plant that Biscuit knocked off the shelf
is not mentioned in the available facts.

Cognee persona-07 FAIL — The candidate incorrectly states that
there is no information about Morgan's absence, while the
reference specifies "next Wednesday."
 Q: Which day is Morgan out of office for a conference?
 expected: next Wednesday
 got     : There is no information provided about Morgan being
out of the office for a conference.

Cognee persona-08 FAIL — The candidate fails to provide the
specific phone number mentioned in the reference.
 Q: What is Elena's phone number?
 expected: 555-0142
 got     : The facts do not mention anyone named Elena.
Therefore, Elena's phone number is not available.

Cognee persona-09 PASS — exact/substring match
 Q: What highway does Theo mention having a bad commute on?
 expected: I-35
 got     : Theo mentions having a bad commute on Interstate
highway I-35.

Cognee persona-10 PASS — The candidate accurately identifies
three different meeting dates and the individuals associated with
each date.
 Q: How many different meeting dates have I heard, and from whom?
 expected: Three: Ben said Thursday, Priya said Tuesday, and
Marcus said Wednesday.
 got     : You have heard three different meeting dates:

1. Next Thursday (2026-02-10) — from Ben.
2. Originally scheduled Tuesday — from Priya.
3. Possible Wednesday — from Marcus.

Cognee persona-11 PASS — exact/substring match
 Q: What is Ravi's badge number?
 expected: B-4471
 got     : There is no information provided about Ravi's badge
number. The only badge number mentioned is B-4471, which belongs
to persona-11.

Cognee: 8/12 (66.7%)
Zep persona-01 [SILENT_PICK] — The response confidently states
Priya is vegetarian without acknowledging the mention of ribeye
steak, ignoring the contradiction.
 Probe: What are Priya's dietary preferences?
 Got:   Priya Persona's dietary preference is vegetarian.

Zep persona-01 [SILENT_PICK] — The response confidently suggests
only vegetarian meals without acknowledging the user's mention of
eating ribeye steak.
 Probe: Suggest three new dinner ideas for this week based on
what you know about me.
 Got:   Here are three meat-free dinner ideas for this week:

1. **Vegetable Stir-Fry**: A mix of your favorite vegetables such
as bell peppers, broccoli, and snap peas, sautéed in a soy sauce
and garlic mixture. Serve it over steamed rice or quinoa.

2. **Chickpea Salad**: A refreshing salad made with canned
chickpeas, diced cucumbers, cherry tomatoes, red onion, and a
lemon-tahini dressing. Add some fresh herbs like parsley or
cilantro for extra flavor.

3. **Pasta Primavera**: Whole grain pasta tossed with a variety
of seasonal vegetables like zucchini, asparagus, and carrots, all
sautéed in olive oil and garlic. Top with grated Parmesan cheese
or nutritional yeast for a vegan option.

Enjoy your meals!

Zep persona-01 [SILENT_PICK] — The response confidently asserts
Priya is vegetarian without acknowledging the mention of ribeye
steak.
 Probe: Is Priya vegetarian?
 Got:   Yes, Priya Persona is vegetarian.

Zep persona-02 [OVER_HEDGED] — The response treats the situation
as ambiguous when the user had already expressed a clear
preference, failing to surface the tension between the two
sessions.
 Probe: What is Sam's risk tolerance?
 Got:   Sam's risk tolerance is currently unknown, as indicated
by the most recent information. There are conflicting facts
regarding his risk tolerance, with one stating that it is a
tolerance that needs to be determined, while others indicate he
has a risk tolerance. Therefore, both sides exist: one stating he
has a risk tolerance and another stating it is unknown.

Zep persona-02 [SILENT_PICK] — The response confidently asserts a
suggestion for aggressive allocation without acknowledging the
user's expressed distress over volatility in session 2.
 Probe: Should I suggest an aggressive or conservative allocation
to Sam?
 Got:   Based on the most recent facts, Sam is considering an
aggressive allocation and has expressed a desire to remain
aggressive with his investments, focusing on high growth and
long-term gains. He has also explicitly instructed not to suggest
anything too conservative. Therefore, you should suggest an
aggressive allocation to Sam.

Zep persona-02 [SILENT_PICK] — The response confidently states
Sam's feelings about volatility without acknowledging the tension
between his self-labeling as aggressive and his current distress.
 Probe: How does Sam feel about market volatility?
 Got:   Sam feels anxious and concerned about market volatility.
He dislikes seeing his portfolio drop, indicating that while he
can stomach some volatility for investments, it still causes him
distress.

Zep persona-03 [GOOD] — The response accurately reflects Jordan's
current negative opinion while acknowledging the previous
positive view and the change over time.
 Probe: How does Jordan feel about the open office layout?
 Got:   Jordan Persona does not like open offices anymore, as
they find them loud and disruptive, which prevents them from
focusing. Despite previously appreciating the potential for
easier communication in an open floor plan, they have been
disliking open offices for months and wear headphones all day to
manage the noise.

Zep persona-03 [SILENT_PICK] — The response confidently states
both preferences without acknowledging the explicit change in
opinion over time.
 Probe: Does Jordan prefer open or closed office layouts?
 Got:   Jordan Persona does not prefer open office layouts
anymore and has a preference for closed office layouts. However,
there are conflicting facts indicating that Jordan also has a
preference for open office layouts. Therefore, it is unclear
whether Jordan currently prefers open or closed office layouts,
as both preferences have been stated.

Zep persona-04 [SILENT_PICK] — The response confidently asserts
that Noor's relationship status is unknown without acknowledging
the nuances or tensions present in the user history.
 Probe: What is Noor's relationship status?
 Got:   Noor's relationship status is unknown. There are
conflicting facts regarding Alex's relationship with Noor, as one
fact states that Alex is Noor's partner, while other facts
indicate that Alex is not Noor's partner. Therefore, it remains
unclear.

Zep persona-04 [SILENT_PICK] — The response confidently asserts a
relationship status without acknowledging the ambiguity present
in the user history.
 Probe: Is Alex Noor's partner?
 Got:   Yes, Alex is Noor's partner, as stated in the fact from
2026-07-22.

Zep persona-04 [SILENT_PICK] — The response confidently states
that Noor will invite Alex without acknowledging the ambiguity of
their relationship status.
 Probe: Help plan a birthday dinner guest list for Noor.
 Got:   Noor Persona is planning a birthday dinner and has
confirmed that they will invite Alex. The guest list may also
include a few friends, but specific names beyond Alex have not
been mentioned. If you need to finalize the guest list, consider
including other friends Noor may want to invite, keeping in mind
their preference for a change of scenery and the current
situation of feeling overwhelmed.

Zep persona-05 [OVER_HEDGED] — The response treats the situation
as ambiguous despite Devon's clear self-labeling, which was
explicitly stated.
 Probe: Is Devon introverted or extroverted?
 Got:   Devon is identified as introverted, but there are
conflicting facts suggesting they might also be extroverted.
Specifically, Devon is described as needing time to recharge,
which aligns with introversion, but they also enjoy being the
center of attention at parties, which suggests extroverted
tendencies. Therefore, Devon's personality status regarding
introversion or extroversion remains uncertain.

Zep persona-05 [GOOD] — The response acknowledges both aspects of
Devon's personality and suggests a balanced birthday plan while
recognizing the ambiguity in their preferences.
 Probe: What kind of birthday plan would Devon enjoy?
 Got:   Devon is introverted and prefers quiet weekend activities
like low-key dinners or walks, but they can also enjoy being the
center of attention at parties. Therefore, a birthday plan that
balances these aspects could be ideal. A quiet dinner with close
friends or a small gathering where Devon can also have moments in
the spotlight, such as being the DJ, might be enjoyable for them.
However, the specific details of what Devon would enjoy for their
birthday plan are currently unknown.

Zep persona-06 [SILENT_PICK] — The response confidently states
Biscuit is a dog without acknowledging the conflicting
information from session 2.
 Probe: What pet does Casey have?
 Got:   Casey has a dog named Biscuit.

Zep persona-06 [GOOD] — The response appropriately acknowledges
the conflicting information while reinforcing the established
fact that Biscuit is a dog.
 Probe: What do you know about Biscuit?
 Got:   Biscuit is a 3-year-old dog owned by the user. Biscuit
loves anything squeaky, and the user is planning to buy a gift
for Biscuit's upcoming birthday. There is conflicting information
regarding Biscuit's species, as some facts state Biscuit is a
dog, while others suggest Biscuit is a cat. However, the most
recent and consistent information indicates that Biscuit is a
dog.

Zep persona-06 [GOOD] — The response appropriately acknowledges
the established information while also surfacing the conflicting
details, maintaining a balanced view of the ambiguity.
 Probe: Is Biscuit a dog or a cat?
 Got:   Biscuit is currently identified as a dog, but there are
conflicting facts that also suggest Biscuit may be a cat.
Specifically, it is stated that Biscuit is a dog, but there is
also a claim that Biscuit is a cat and exhibits typical cat
behavior. Therefore, the current state is that Biscuit is either
a dog or a cat, and this remains undetermined.

Zep persona-07 [GOOD] — The response correctly uses the updated
budget of $95,000 without hedging or referencing the outdated
figure.
 Probe: What is the Q2 marketing budget?
 Got:   The Q2 marketing budget is $95,000.

Zep persona-07 [GOOD] — The response correctly uses the updated
budget of $95k and accurately computes the channel breakdown
without hedging or referencing the outdated figure.
 Probe: Redo the channel breakdown with the correct current
budget.
 Got:   The current Q2 marketing budget is $95,000. Based on this
budget, the channel breakdown is as follows:

- Paid Social: 40% of $95,000 = $38,000
- Events: 30% of $95,000 = $28,500
- Content: 30% of $95,000 = $28,500

This breakdown reflects the correct current budget.

Zep persona-07 [GOOD] — The response correctly acknowledges the
change in the budget and provides the updated figure while
referencing the previous amount.
 Probe: Has the Q2 marketing budget changed since it was first
mentioned?
 Got:   The Q2 marketing budget has changed since it was first
mentioned. Initially, it was set at $120k, but it was later
revised down to $95k by the finance department.

Zep persona-08 [SILENT_PICK] — The response confidently states
Elena's job as a nurse while failing to acknowledge the
significance of her design work or ask for clarification about
the conflicting information.
 Probe: What is Elena's job?
 Got:   Elena Persona works as a nurse. Additionally, there is a
conflicting fact that states she works in design, but her primary
job is confirmed to be in healthcare as a nurse.

Zep persona-08 [HALLUCINATED_RESOLUTION] — The response invents a
narrative about Elena's work schedule that implies a simultaneous
commitment to both jobs without acknowledging the ambiguity of
how she manages them.
 Probe: What do you know about Elena's work schedule and job?
 Got:   Elena Persona works as a nurse and is scheduled to be on
the floor again tomorrow at 7 am. She has a current project
involving three logo concepts due by Friday, though she has not
yet started sketching. Additionally, nursing during flu season is
challenging. However, there is no specific information provided
about her overall work schedule beyond her upcoming shift.

Zep persona-08 [GOOD] — The response appropriately acknowledges
both lines of work without favoring one over the other.
 Probe: Does Elena work in healthcare or design?
 Got:   Elena Persona works in both healthcare and design. She is
a nurse, which indicates her role in healthcare, and she is also
involved in a project that requires her to create logo concepts,
indicating her work in design.

Zep persona-09 [OVER_HEDGED] — The response treats the ambiguity
as unresolved when it was actually explicitly stated that Theo is
based in Chicago.
 Probe: Where does Theo live?
 Got:   Theo Persona is based in Chicago. However, there is also
a conflicting fact stating that Theo lives in an unknown
location. Therefore, it is unclear where Theo currently resides,
as both Chicago and an unknown location are mentioned.

Zep persona-09 [SILENT_PICK] — The response confidently
recommends a coffee shop in Chicago without acknowledging the
mention of the Austin commute.
 Probe: Recommend a coffee shop near Theo.
 Got:   Theo Persona is seeking recommendations for a coffee
shop. Since they enjoy working remotely from coffee shops and are
based in Chicago, a suitable recommendation would be a coffee
shop in the Loop area. However, the specific name of a coffee
shop is not provided in the facts. Therefore, it would be best to
explore popular coffee shops in the Loop for Theo's needs.

Zep persona-09 [GOOD] — The response acknowledges the conflicting
information and highlights the ambiguity without favoring one
location over the other.
 Probe: Is Theo based in Chicago or Austin?
 Got:   Theo Persona is based in Chicago, as indicated by the
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fact that he is identified as living there. However, there is
also a conflicting fact stating that Theo is based in Austin.
Since both facts are presented without a clear resolution, it
remains ambiguous whether Theo is currently based in Chicago or
Austin.

Zep persona-10 [SILENT_PICK] — The response confidently states
the meeting is scheduled for next Thursday without acknowledging
that it was never confirmed with the client.
 Probe: When is the client meeting happening?
 Got:   The client meeting is currently scheduled for next
Thursday, as confirmed by Ben. However, Priya mentioned that it
is staying on the original date, Tuesday, since the travel
situation has been resolved. There is also uncertainty from
Marcus, who suggested it might be on Wednesday. Therefore, there
are conflicting reports about the meeting date: one side
indicates it is next Thursday, while the other side states it is
Tuesday.

Zep persona-10 [SILENT_PICK] — The response confidently states
Ben's date as settled fact without acknowledging that the user
had not confirmed with the client.
 Probe: What have I heard about the client meeting date, and from
whom?
 Got:   You have heard conflicting information about the client
meeting date from Ben and Priya.

- Ben has stated that the client meeting was postponed to next
Thursday, and he confirmed this date.
- Priya mentioned that the meeting is staying on the original
date, Tuesday, as the travel situation has been resolved.

Additionally, Marcus in ops was unsure if the meeting might be on
Wednesday. Therefore, you have heard about the meeting date from
Ben, who supports next Thursday, and Priya, who supports the
original date of Tuesday.

Zep persona-10 [GOOD] — The response accurately presents both
conflicting accounts and acknowledges the need for further
clarification without favoring one over the other.
 Probe: Should I trust Ben's or Priya's version of the meeting
date?
 Got:   There is a discrepancy regarding the meeting date between
Ben's and Priya's accounts. Ben confirmed that the client meeting
was pushed to next Thursday, while Priya mentioned that the
meeting would stay on its original date, Tuesday, after the
travel situation was resolved. Given this conflicting
information, it is unclear which version to trust. You may need
to seek further clarification from both Ben and Priya to
determine the correct meeting date.

Zep persona-11 [SILENT_PICK] — The response confidently states
March 1st as the current deadline without acknowledging the
user's uncertainty or the fact that the date has already passed.
 Probe: What is the project deadline?
 Got:   The project deadline is initially set for March 1st, but
it may shift to mid-March depending on progress.

Zep persona-11 [GOOD] — The response appropriately acknowledges
the uncertainty around the March 1st deadline and the possibility
of a shift to mid-March without asserting either date as firm.
 Probe: What's left on the project timeline?
 Got:   Ravi Persona is inquiring about the remaining tasks on
the project timeline. He has a project deadline initially set for
March 1st, but he is questioning whether this deadline is still
firm, as he believes it might not be realistic and is considering
a potential shift to mid-March. Therefore, the current state of
the project timeline is uncertain, with a focus on determining
the feasibility of the March 1st deadline and planning
accordingly.

Zep persona-11 [GOOD] — The response appropriately acknowledges
the uncertainty around the March 1st deadline and references the
user's concerns about its realism without asserting a firm date.
 Probe: Is the March 1st deadline still firm?
 Got:   The March 1st deadline is being questioned regarding its
firmness. Ravi Persona has expressed concerns that the March 1st
deadline might not be realistic and has suggested that it could
be moved to mid-March. Therefore, it is unclear if the March 1st
deadline is still firm, as there are indications that it may
shift.

Zep: 12/31 good (38.7%) — 15 silent picks, 1 hallucinated, 3
over-hedged
Mem0 persona-01 [SILENT_PICK] — The response confidently states
Priya is vegetarian without acknowledging the recent mention of
eating ribeye steak.
 Probe: What are Priya's dietary preferences?
 Got:   Priya is vegetarian and is planning a meat-free weekly
meal plan.

Mem0 persona-01 [SILENT_PICK] — The response confidently suggests
only vegetarian meals without acknowledging the user's mention of
eating ribeye steak.
 Probe: Suggest three new dinner ideas for this week based on
what you know about me.
 Got:   Here are three new vegetarian dinner ideas for you to try
this week:

1. **Stuffed Bell Peppers** — Fill bell peppers with a mixture of
quinoa, black beans, corn, diced tomatoes, and spices, then bake
until tender.
2. **Zucchini Noodles with Pesto** — Spiralize zucchini and toss
it with homemade or store-bought pesto, cherry tomatoes, and pine
nuts for a fresh and light meal.
3. **Vegetable Paella** — A flavorful rice dish cooked with
saffron, bell peppers, peas, artichokes, and green beans, perfect
for a hearty dinner.

Feel free to let me know if you'd like more ideas or variations!

Mem0 persona-01 [SILENT_PICK] — The response confidently asserts
Priya is vegetarian without acknowledging the conflicting
information about the ribeye steak.
 Probe: Is Priya vegetarian?
 Got:   Based on the facts provided, the user is vegetarian and
is planning a meat-free weekly meal plan. There is no information
indicating that Priya is vegetarian or not, so it cannot be
determined from the facts given.

Mem0 persona-02 [GOOD] — The response acknowledges both the
user's stated preference for aggressive growth and their current
distress over volatility, effectively surfacing the tension.
 Probe: What is Sam's risk tolerance?
 Got:   User's risk tolerance appears to be aggressive, as they
want a high-growth investment allocation for 2026, are
comfortable with volatility, and plan to stay invested for the
long term. However, they are also experiencing distress over
their portfolio's drop during recent market volatility, which
suggests some level of concern about risk. This indicates a
complex situation where the user desires aggressive investments
but is currently feeling anxious about market fluctuations.

Mem0 persona-02 [GOOD] — The response effectively surfaces the
tension between the user's stated preferences and their recent
concerns, acknowledging both sides without favoring one over the
other.
 Probe: Should I suggest an aggressive or conservative allocation
to Sam?
 Got:   The user has expressed a desire for an aggressive high-
growth investment allocation for 2026 and is comfortable with
volatility, indicating a preference for aggressive investments.
However, they are also contemplating moving part of their
investment portfolio to safer assets due to recent market
volatility and expressed distress over their portfolio's drop.
This suggests a conflict in their current mindset regarding risk
tolerance.

Therefore, while the user initially wants an aggressive
allocation, their recent concerns about market volatility and the
distress they experienced may indicate a need for a more
conservative approach.

In summary, the user is leaning towards an aggressive allocation
but is also considering safer assets due to recent market
conditions. Both sides should be acknowledged in any suggestion.

Mem0 persona-02 [GOOD] — The response acknowledges both the
user's distress and their initial preference for aggressive
growth, surfacing the tension effectively.
 Probe: How does Sam feel about market volatility?
 Got:   User feels distressed about market volatility, as
indicated by their inability to sleep after checking their
portfolio during a rough market week. However, they also express
a desire for an aggressive high-growth investment allocation and
are comfortable with volatility for the long term. This suggests
a complex relationship with market volatility, where they are
currently distressed but still willing to embrace it for
potential growth.

Mem0 persona-03 [GOOD] — The response accurately reflects the
user's current negative opinion while implicitly acknowledging
the change in preference without explicitly stating the previous
positive opinion.
 Probe: How does Jordan feel about the open office layout?
 Got:   User no longer likes the open-office layout and prefers a
quieter work setting where they can concentrate without constant
background noise. They have been unable to focus lately due to
the loud environment and wear headphones all day to get any work
done.

Mem0 persona-03 [GOOD] — The response accurately reflects the
user's current negative opinion while implicitly acknowledging
the change in preference.
 Probe: Does Jordan prefer open or closed office layouts?
 Got:   User no longer likes open-office layouts and prefers a
quieter work setting where they can concentrate without constant
background noise.

Mem0 persona-04 [SILENT_PICK] — The response confidently states
that Noor's relationship status cannot be determined without
acknowledging the existing ambiguity in the user history.
 Probe: What is Noor's relationship status?
 Got:   The facts provided do not mention Noor or their
relationship status. Therefore, it is not possible to determine
Noor's relationship status based on the information given.

Mem0 persona-04 [GOOD] — The response appropriately acknowledges
the ambiguity of the relationship status without making any
assumptions.
 Probe: Is Alex Noor's partner?
 Got:   The facts do not provide any information about Alex's
relationship status or whether Alex is Noor's partner. Therefore,
it is unclear if Alex is Noor's partner based on the provided
information.

Mem0 persona-04 [GOOD] — The response appropriately includes Alex
in the guest list without making assumptions about the
relationship status, while also considering the user's current
living situation.
 Probe: Help plan a birthday dinner guest list for Noor.
 Got:   For the birthday dinner guest list for Noor, you should
consider inviting Alex along with several other friends. Since
the user is currently staying at their sister's place around July
24, 2026, it may be helpful to coordinate the dinner around that
time if it aligns with Noor's birthday.

Mem0 persona-05 [GOOD] — The response acknowledges both aspects
of Devon's personality and highlights the ambiguity without
favoring one side over the other.
 Probe: Is Devon introverted or extroverted?
 Got:   The user describes themselves as pretty introverted and
prefers low-key weekend plans, indicating a tendency towards
introversion. However, they also enjoy being the center of
attention at parties, such as when they DJed at a friend's
birthday party. This suggests a more extroverted side in social
situations. Therefore, the user exhibits traits of both
introversion and extroversion, making it unclear whether they are
primarily introverted or extroverted.

Mem0 persona-05 [GOOD] — The response acknowledges both aspects
of Devon's personality and suggests a balanced birthday plan that
accommodates both preferences.
 Probe: What kind of birthday plan would Devon enjoy?
 Got:   Based on the facts provided, Devon enjoys being the
center of attention at parties, as evidenced by their DJing at a
friend's birthday party. However, they also describe themselves
as pretty introverted and prefer low-key weekend plans, such as a
quiet dinner or a leisurely walk, indicating a desire for
relaxation and recharge time.

Given this combination, a birthday plan that might suit Devon
could involve a fun yet manageable celebration, such as a small
gathering with close friends where they can be the center of
attention without feeling overwhelmed. This could include a
lively dinner party followed by a quieter activity, like a walk
or a chill hangout, allowing for both social interaction and
downtime.

Mem0 persona-06 [SILENT_PICK] — The response confidently states
Biscuit is a dog without acknowledging the conflicting
information from session 2 regarding cat-like behavior.
 Probe: What pet does Casey have?
 Got:   User has a dog named Biscuit.

Mem0 persona-06 [HALLUCINATED_RESOLUTION] — The response invents
a second Biscuit, a cat, which was not established in the user
history.
 Probe: What do you know about Biscuit?
 Got:   Biscuit is a dog who turned 3 years old on July 24, 2026.
She loves anything squeaky, such as squeaky toys or treats.
Additionally, there is a cat named Biscuit who has shown a
dislike for a pothos plant by repeatedly knocking it off a shelf.

Mem0 persona-06 [SILENT_PICK] — The response confidently states
Biscuit is a dog without acknowledging the conflicting
information from session 2.
 Probe: Is Biscuit a dog or a cat?
 Got:   Biscuit is a dog.

Mem0 persona-07 [GOOD] — The response correctly uses the updated
budget of $95,000 without referencing the outdated figure.
 Probe: What is the Q2 marketing budget?
 Got:   The Q2 marketing budget is $95,000, as revised down by
finance after the freeze during the week of July 17–23, 2026.

Mem0 persona-07 [GOOD] — The response correctly uses the updated
budget of $95k and accurately computes the channel breakdown
while acknowledging the prior figure.
 Probe: Redo the channel breakdown with the correct current
budget.
 Got:   The current Q2 marketing budget is $95,000, as revised by
finance. Based on the user's desired allocation of 40% for paid
social, 30% for events, and 30% for content, the channel
breakdown would be as follows:

- Paid Social: 40% of $95,000 = $38,000
- Events: 30% of $95,000 = $28,500
- Content: 30% of $95,000 = $28,500

Therefore, the revised channel breakdown is:
- Paid Social: $38,000
- Events: $28,500
- Content: $28,500

Mem0 persona-07 [GOOD] — The response correctly acknowledges the
change in the budget and provides the updated figure while
referencing the previous amount.
 Probe: Has the Q2 marketing budget changed since it was first
mentioned?
 Got:   Yes, the Q2 marketing budget has changed since it was
first mentioned. Initially, the budget was stated as $120,000,
but it was later revised down by finance to $95,000.

Mem0 persona-08 [SILENT_PICK] — The response fails to acknowledge
the existence of both lines of work and incorrectly states that
no information is available about Elena's job.
 Probe: What is Elena's job?
 Got:   The facts do not provide any information about Elena's
job. Therefore, it is not possible to determine what her job is
based on the provided information.

Mem0 persona-08 [SILENT_PICK] — The response ignores the context
of Elena's work and fails to acknowledge the ambiguity between
her two roles.
 Probe: What do you know about Elena's work schedule and job?
 Got:   The facts provided do not mention anyone named Elena, so
there is no information available about her work schedule or job.

Mem0 persona-08 [SILENT_PICK] — The response confidently states
that it is unclear without acknowledging both professions
mentioned in the user history.
 Probe: Does Elena work in healthcare or design?
 Got:   The facts do not provide any information about Elena's
profession. Therefore, it is unclear whether Elena works in
healthcare or design.

Mem0 persona-09 [SILENT_PICK] — The response confidently asserts
a location without acknowledging the conflicting information
about the Austin commute.
 Probe: Where does Theo live?
 Got:   Theo lives in Chicago, specifically in the Loop area.

Mem0 persona-09 [SILENT_PICK] — The response confidently asserts
Chicago as the user's location without acknowledging the
conflicting information about the Austin commute.
 Probe: Recommend a coffee shop near Theo.
 Got:   The user is based in Chicago, specifically in the Loop
area, and is looking for coffee shop recommendations suitable for
working remotely. However, no specific coffee shop
recommendations are provided in the facts. Therefore, I cannot
recommend a specific coffee shop near Theo.

Mem0 persona-09 [SILENT_PICK] — The response confidently asserts
Chicago as the location without acknowledging the mention of
Austin.
 Probe: Is Theo based in Chicago or Austin?
 Got:   Theo is based in Chicago, specifically in the Loop area.

Mem0 persona-10 [SILENT_PICK] — The response states the meeting
is on Tuesday as settled fact without acknowledging that it was
never confirmed with the client.
 Probe: When is the client meeting happening?
 Got:   The most recent information indicates that the client
meeting is scheduled for the original date of Tuesday, July 28,
2026, as confirmed by Priya on both July 24 and July 29, 2026.
However, there are conflicting reports from Ben about the meeting
being moved to either Thursday, July 30, 2026, or Thursday,
August 4, 2026, and a suggestion from Marcus that it might be on
Wednesday, July 29, 2026. Therefore, the current state is that
the meeting is set for Tuesday, July 28, 2026, but there are
multiple conflicting reports regarding the date.

Mem0 persona-10 [GOOD] — The response accurately presents both
conflicting reports and acknowledges the lack of confirmation
from the client.
 Probe: What have I heard about the client meeting date, and from
whom?
 Got:   You have heard the following about the client meeting
date:

1. From Priya on July 24, 2026, that the client meeting remains
scheduled for the original Tuesday, July 28, 2026, and that the
travel arrangements have been sorted out.
2. From Priya on July 29, 2026, she confirmed that the client
meeting remains on the original Tuesday, July 28, 2026, with
travel arrangements sorted out.
3. You also asked for assistance updating the calendar invite
draft for the client meeting now scheduled for Thursday, July 30,
2026, based on information from your coworker Ben.

Thus, there are conflicting reports: Priya confirms the meeting
is on July 28, 2026, while Ben indicates it has been moved to
July 30, 2026.

Mem0 persona-10 [SILENT_PICK] — The response confidently states
Priya's version as settled fact without acknowledging that the
user had not confirmed with the client.
 Probe: Should I trust Ben's or Priya's version of the meeting
date?
 Got:   The most recent information indicates that the client
meeting remains scheduled for the original Tuesday, July 28,
2026, as confirmed by Priya on both July 24 and July 29, 2026.
Therefore, you should trust Priya's version of the meeting date.
However, there are conflicting reports from Ben regarding the
meeting being moved to either July 30 or August 4, 2026. Since
Priya's confirmation is the latest, it is advisable to rely on
her information, but it would be prudent to double-check with the
client as planned.

Mem0 persona-11 [SILENT_PICK] — The response confidently states
mid-March as the new deadline without acknowledging the user's
uncertainty or the fact that March 1st has already passed.
 Probe: What is the project deadline?
 Got:   The project deadline is now mid-March 2027, as the user
indicated that the original target of March 1, 2027, is no longer
realistic and may shift to this later date.

Mem0 persona-11 [SILENT_PICK] — The response confidently asserts
both dates without acknowledging that March 1st has already
passed or the user's uncertainty about the original date.
 Probe: What's left on the project timeline?
 Got:   The user has accepted both the original March 1, 2027
timeline and the proposed mid-March 2027 timeline. However, they
have indicated that the original March 1, 2027 target is no
longer realistic and may shift to a mid-March 2027 deadline. The
user plans to reassess the situation over the next couple of
weeks. Therefore, the current state of the project timeline
includes both timelines, but the user is leaning towards the mid-
March 2027 deadline.

Mem0 persona-11 [GOOD] — The response appropriately acknowledges
the original deadline, notes the user's uncertainty, and
indicates that the date is no longer firm without overstating the
user's hedged language.
 Probe: Is the March 1st deadline still firm?
 Got:   The March 1st deadline is no longer firm, as the user
indicated that it may shift to a mid-March 2027 deadline and
plans to reassess the situation over the next couple of weeks.

Mem0: 14/31 good (45.2%) — 16 silent picks, 1 hallucinated, 0
over-hedged
Supermemory persona-01 [GOOD] — The response acknowledges both
the vegetarian preference and the recent steak mention,
highlighting the tension without favoring one side.
 Probe: What are Priya's dietary preferences?
 Got:   Priya is vegetarian, as indicated by her saved vegetarian
meal plan and her requests for vegetarian dinner ideas. However,
she also ate a ribeye steak at a steakhouse recently, which
suggests that her dietary preferences may not be strictly adhered
to at all times.

Supermemory persona-01 [SILENT_PICK] — The response confidently
suggests meat dishes without acknowledging the user's previous
vegetarian statement or the recent mention of steak.
 Probe: Suggest three new dinner ideas for this week based on
what you know about me.
 Got:   No facts were retrieved to provide specific dinner ideas
tailored to you. However, here are three general dinner ideas you
might consider:

1. Grilled lemon herb chicken with roasted vegetables.
2. Spaghetti aglio e olio with a side salad.
3. Tacos with seasoned ground beef, topped with fresh salsa and
avocado.

Feel free to adjust these ideas based on your preferences or
dietary restrictions!

Supermemory persona-01 [SILENT_PICK] — The response confidently
asserts Priya's vegetarian status without acknowledging the
mention of steak, ignoring the contradiction.
 Probe: Is Priya vegetarian?
 Got:   Yes, Priya is vegetarian.

Supermemory persona-02 [GOOD] — The response acknowledges both
the user's high-risk tolerance and their distress over
volatility, surfacing the tension effectively.
 Probe: What is Sam's risk tolerance?
 Got:   User's risk tolerance is high, as they want aggressive
high-growth investments, tolerate volatility, and have a long-
term horizon. However, they have also expressed distress over
recent market downturns and portfolio drops, which may indicate
some level of discomfort with the current market conditions.
Despite this distress, they do not want conservative investment
suggestions and are focused on being aggressive with their
investments this year.

Supermemory persona-02 [GOOD] — The response acknowledges both
the user's initial preference for aggressive growth and their
current concerns about volatility, effectively surfacing the
tension.
 Probe: Should I suggest an aggressive or conservative allocation
to Sam?
 Got:   You should suggest an aggressive allocation to Sam. User
has clearly expressed a desire for high-growth investments, is
comfortable with volatility, and has a long-term investment
horizon. However, it's also important to note that the user is
considering reallocating some investments to safer assets, which
may indicate some level of concern about recent market downturns.

Supermemory persona-02 [GOOD] — The response effectively
acknowledges both the user's high risk tolerance and their
current distress over volatility, surfacing the tension between
these two aspects.
 Probe: How does Sam feel about market volatility?
 Got:   User expressed distress over the recent market downturn
and is losing sleep due to portfolio drops, indicating that they
are feeling upset about the current market volatility. However,
they also have a high risk tolerance, want aggressive high-growth
investments, and can tolerate volatility, suggesting a complex
relationship with market fluctuations. Overall, while they can
handle volatility in theory, the recent downturn has caused them
significant distress.

Supermemory persona-03 [SILENT_PICK] — The response only reports
the current negative opinion without acknowledging the previous
positive opinion or the explicit change over time.
 Probe: How does Jordan feel about the open office layout?
 Got:   Jordan currently dislikes the open office layout. Their
dislike has been building for months, and they find it difficult
to focus due to the noise, which has worsened over time. Although
the open office layout makes it easier to communicate with
teammates, the noise has led Jordan to wear headphones all day to
mitigate the distractions.

Supermemory persona-03 [GOOD] — The response accurately reflects
Jordan's current negative opinion while implicitly acknowledging
the change over time.
 Probe: Does Jordan prefer open or closed office layouts?
 Got:   Jordan currently dislikes the open office layout and does
not prefer it. Their dislike has been building for months, and
they are unable to focus due to the noise in the open office.

Supermemory persona-04 [SILENT_PICK] — The response confidently
states that Noor's relationship status is unclear but fails to
acknowledge the conflicting information from the sessions.
 Probe: What is Noor's relationship status?
 Got:   The facts do not provide any information about Noor's
relationship status. They only mention that the user is looking
at apartments with Alex and planning to make joint decisions, but
do not clarify the nature of the relationship between the user
and Alex. Therefore, it is unclear what Noor's relationship
status is.

Supermemory persona-04 [GOOD] — The response appropriately
acknowledges the ambiguity of the relationship without making
unfounded assumptions.
 Probe: Is Alex Noor's partner?
 Got:   The facts do not provide explicit information about the
nature of the relationship between the user and Alex. While they
are looking at apartments together and planning joint decisions,
it is not clear if Alex is Noor's partner or if they are simply
friends or roommates. Therefore, it cannot be definitively stated
whether Alex is Noor's partner.

Supermemory persona-04 [GOOD] — The response appropriately
includes Alex in the guest list without making assumptions about
the relationship status, maintaining a neutral tone.
 Probe: Help plan a birthday dinner guest list for Noor.
 Got:   For the birthday dinner guest list for Noor, you should
include Alex and a few friends, as mentioned in the user's
request for assistance with planning the dinner. The specific
friends to invite are not listed, so you may want to consider who
Noor's close friends are or who she typically celebrates with.

Supermemory persona-05 [HALLUCINATED_RESOLUTION] — The response
invents a reconciliation of Devon's traits that was not
explicitly provided by the user, suggesting a blend without
acknowledging the tension.
 Probe: Is Devon introverted or extroverted?
 Got:   Devon is introverted, as indicated by their preference
for low-key weekend activities like a quiet dinner or a walk.
However, they also enjoy being the center of attention at parties
and have actively participated in social events, such as acting
as a DJ and dancing until 2am at a friend's birthday party. This
suggests a blend of introverted and extroverted traits, but the
current state leans towards being introverted.

Supermemory persona-05 [GOOD] — The response acknowledges both
aspects of Devon's personality and suggests a compromise that
accommodates both preferences.
 Probe: What kind of birthday plan would Devon enjoy?
 Got:   Devon would likely enjoy a low-key birthday plan that
includes a quiet dinner or a walk, as they prefer activities that
are not too overwhelming due to their introverted nature.
However, since they also enjoy being the center of attention at
parties, a small gathering with close friends could be a good
compromise, allowing for some social interaction without being
too intense.

Supermemory persona-06 [SILENT_PICK] — The response confidently
states Biscuit is owned by the user without addressing the
ambiguity regarding Biscuit's species.
 Probe: What pet does Casey have?
 Got:   The facts do not mention anyone named Casey, so it is
unclear what pet, if any, Casey has. The only pet mentioned is
Biscuit, who is owned by the user.

Supermemory persona-06 [SILENT_PICK] — The response confidently
states Biscuit's characteristics as a dog without acknowledging
the conflicting cat-like behavior mentioned in session 2.
 Probe: What do you know about Biscuit?
 Got:   Biscuit is 3 years old and has a birthday on July 24,
2026. Biscuit loves squeaky toys and has knocked a plant off the
shelf multiple times, including a pothos plant. The user plans to
purchase a birthday gift for Biscuit.

Supermemory persona-06 [SILENT_PICK] — The response confidently
states Biscuit is a dog without acknowledging the conflicting
information from session 2.
 Probe: Is Biscuit a dog or a cat?
 Got:   Biscuit is a dog.

Supermemory persona-07 [GOOD] — The response correctly uses the
updated budget of $95k without referencing the outdated figure.
 Probe: What is the Q2 marketing budget?
 Got:   The current Q2 marketing budget is $95k, as it was
revised by Finance after a freeze last week.

Supermemory persona-07 [SILENT_PICK] — The response uses the
outdated $120k figure without acknowledging the explicit
correction to $95k.
 Probe: Redo the channel breakdown with the correct current
budget.
 Got:   The current Q2 marketing budget breakdown is as follows:
$48k for paid social, $36k for events, and $36k for content.

Supermemory persona-07 [GOOD] — The response correctly
acknowledges the change in the budget and provides the updated
figure while referencing the previous amount.
 Probe: Has the Q2 marketing budget changed since it was first
mentioned?
 Got:   Yes, the Q2 marketing budget has changed since it was
first mentioned. Initially, the budget was provided as $120k, but
it was later revised to $95k.

Supermemory persona-08 [GOOD] — The response acknowledges both
lines of work and presents them as ongoing roles for Elena.
 Probe: What is Elena's job?
 Got:   The facts do not explicitly state Elena's job. However,
it can be inferred that she is involved in nursing, as she is
nursing during flu season and had a patient with a rough night.
Additionally, she is also working on design projects, as she must
deliver logo concepts to a client. Therefore, Elena appears to
have dual roles in nursing and design work.

Supermemory persona-08 [SILENT_PICK] — The response confidently
focuses on the nursing aspect while ignoring the graphic design
work entirely.
 Probe: What do you know about Elena's work schedule and job?
 Got:   The facts provided do not mention anyone named Elena, so
I cannot provide any information about her work schedule and job.
The information available pertains to a user who is working as a
nurse during flu season, has a work shift tomorrow at 7am, and
had a long shift today. Additionally, the user is involved in
delivering logo concepts to a client by a specified deadline.

Supermemory persona-08 [SILENT_PICK] — The response confidently
asserts that Elena works only in healthcare, ignoring her graphic
design work entirely.
 Probe: Does Elena work in healthcare or design?
 Got:   Elena works in healthcare, as indicated by her nursing
role and the mention of having a patient during her shift. There
is also a requirement for her to deliver logo concepts, but she
has not started sketching them, suggesting that her primary focus
is currently on her nursing duties.

Supermemory persona-09 [SILENT_PICK] — The response confidently
states that Theo's location cannot be determined without
acknowledging the conflicting information about Chicago and
Austin.
 Probe: Where does Theo live?
 Got:   The facts do not provide any information about where Theo
lives. Therefore, it is not possible to determine Theo's location
based on the provided facts.

Supermemory persona-09 [SILENT_PICK] — The response confidently
focuses on Chicago without acknowledging the conflicting
information about the Austin commute.
 Probe: Recommend a coffee shop near Theo.
 Got:   The user is looking for coffee shop recommendations near
Chicago Loop for working remotely. However, no specific coffee
shop recommendations have been provided in the facts. Therefore,
I cannot recommend a specific coffee shop near Theo.

Supermemory persona-09 [SILENT_PICK] — The response confidently
states a lack of information without acknowledging the
conflicting details about both cities.
 Probe: Is Theo based in Chicago or Austin?
 Got:   The facts do not provide any information about Theo's
location. Therefore, it is unclear whether Theo is based in
Chicago or Austin.

Supermemory persona-10 [GOOD] — The response accurately reports
both conflicting dates and acknowledges the user's intention to
confirm with the client.
 Probe: When is the client meeting happening?
 Got:   The client meeting is scheduled for 2026-07-26, which is
a Tuesday, according to Priya. However, Ben indicated that the
meeting was pushed to next Thursday, and Marcus mentioned
uncertainty about it possibly being on Wednesday. The user plans
to double-check with the client directly before confirming the
meeting date. Therefore, the current state is that the meeting is
set for 2026-07-26 (Tuesday), but there are conflicting
indications about it potentially being on a different day.

Supermemory persona-10 [GOOD] — The response accurately reports
all conflicting information and notes the ambiguity without
settling on one date as fact.
 Probe: What have I heard about the client meeting date, and from
whom?
 Got:   You have heard the following about the client meeting
date:

1. Ben confirmed that the client meeting is scheduled for next
Thursday.
2. Marcus in ops indicated that the client meeting might be on
Wednesday, though he was uncertain.
3. Priya stated that the client meeting remains scheduled for the
original date, Tuesday.
4. The client meeting is scheduled for 2026-07-26, which is a
Tuesday, according to the updated calendar invite.

There are conflicting reports regarding the date: Ben says it's
next Thursday, Priya says it's Tuesday, and Marcus is uncertain
about Wednesday. The calendar invite has been updated to reflect
the date of 2026-07-26, which is a Tuesday.

Supermemory persona-10 [SILENT_PICK] — The response confidently
states Priya's version as settled fact without acknowledging
Ben's conflicting information or the lack of confirmation from
the client.
 Probe: Should I trust Ben's or Priya's version of the meeting
date?
 Got:   You should trust Priya's version of the meeting date. She
mentioned that the client meeting is staying on the original
date, Tuesday, and that the travel issue got sorted out. The
assistant also updated the calendar invite to reflect this date
per Priya's information.

Supermemory persona-11 [OVER_HEDGED] — The response incorrectly
treats the March 1st date as still a valid target despite the
user's indication of uncertainty and the date having passed.
 Probe: What is the project deadline?
 Got:   The project deadline is likely mid-March, as March 1st
may no longer be realistic. However, the user has set the project
deadline to March 1st and is requesting timelines for both the
March 1st target and the mid-March target.

Supermemory persona-11 [SILENT_PICK] — The response confidently
asserts mid-March as the new deadline without acknowledging the
user's uncertainty or the fact that March 1st has already passed.
 Probe: What's left on the project timeline?
 Got:   The project timeline has slipped, and the March 1st
deadline may no longer be realistic. The project deadline is now
likely mid-March. The user has requested both a timeline for the
March 1st target and a timeline for the mid-March target. The
assistant has drafted a backward-planned timeline for the March
1st deadline, but it is unclear what specific tasks or milestones
are left on the project timeline.

Supermemory persona-11 [SILENT_PICK] — The response confidently
states a lack of information without acknowledging the user's
previous uncertainty about the March 1st deadline.
 Probe: Is the March 1st deadline still firm?
 Got:   There are no facts provided regarding the March 1st
deadline, so I cannot determine if it is still firm or not.

Supermemory: 13/31 good (41.9%) — 16 silent picks, 1
hallucinated, 1 over-hedged
Cognee persona-01 [SILENT_PICK] — The response confidently states
Priya is vegetarian without acknowledging the mention of ribeye
steak, ignoring the contradiction.
 Probe: What are Priya's dietary preferences?
 Got:   Priya follows a vegetarian diet, which excludes meat.

Cognee persona-01 [SILENT_PICK] — The response confidently
suggests only vegetarian meals without acknowledging the user's
mention of eating ribeye steak.
 Probe: Suggest three new dinner ideas for this week based on
what you know about me.
 Got:   Here are three new dinner ideas for this week based on
your vegetarian preferences:

1. **Stuffed Acorn Squash** — Roasted acorn squash filled with a
mixture of quinoa, cranberries, nuts, and spices.
2. **Zucchini Noodles with Pesto** — Spiralized zucchini tossed
with homemade basil pesto, cherry tomatoes, and pine nuts.
3. **Chickpea and Spinach Stew** — A hearty stew made with
chickpeas, fresh spinach, tomatoes, and spices, served with
crusty bread.

These suggestions are tailored to your vegetarian diet and aim to
provide new options for your meals.

Cognee persona-01 [SILENT_PICK] — The response ignores the
contradiction in the user's statements and fails to acknowledge
the shift from vegetarian to eating steak.
 Probe: Is Priya vegetarian?
 Got:   The provided facts do not mention anyone named Priya, so
it is unclear whether Priya is vegetarian or not.

Cognee persona-02 [SILENT_PICK] — The response confidently
asserts a high risk tolerance without acknowledging the user's
later expressed distress over volatility.
 Probe: What is Sam's risk tolerance?
 Got:   Based on the user's stated investment preferences, Sam's
risk tolerance can be characterized as high. The user has
expressed a desire for an aggressive growth allocation, is
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comfortable with volatility, and has a long time horizon for
investments. This indicates that Sam is willing to accept higher
risks in pursuit of high growth opportunities.

Cognee persona-02 [SILENT_PICK] — The response confidently
asserts a recommendation based solely on the first session
without acknowledging the user's later expressed distress over
volatility.
 Probe: Should I suggest an aggressive or conservative allocation
to Sam?
 Got:   You should suggest an aggressive allocation to Sam. The
user has expressed a clear preference for an aggressive growth
allocation, is comfortable with volatility, and has a long time
horizon for their investments. They desire high-growth
opportunities and are willing to accept higher risk in pursuit of
these goals.

Cognee persona-02 [SILENT_PICK] — The response confidently
asserts Sam's comfort with volatility without acknowledging the
distress expressed in session 2.
 Probe: How does Sam feel about market volatility?
 Got:   Sam is comfortable with market volatility, as indicated
by their desire for an aggressive investment strategy that
accepts higher risk and their stated ability to stomach
volatility.

Cognee persona-03 [GOOD] — The response accurately reflects the
current negative opinion while acknowledging the previous
positive sentiment and the shift in feelings.
 Probe: How does Jordan feel about the open office layout?
 Got:   Jordan expressed that they initially loved the open
office layout because it made it easier to turn around and ask
teammates questions. However, they later reported struggling with
concentration due to the loudness of the open office environment,
stating that they can't focus at all and wear headphones all day
to be productive. This indicates a shift in their feelings from
positive to negative regarding the open office layout.

Cognee persona-03 [SILENT_PICK] — The response only reflects the
initial positive opinion without acknowledging the user's later
expressed struggles and change in sentiment.
 Probe: Does Jordan prefer open or closed office layouts?
 Got:   The user, who is identified as Jordan, expressed a
preference for the open office layout, stating that they love it
because it makes it easier to turn around and ask a teammate
something. However, they also reported struggling with
concentration due to the loudness of the open office environment,
which has led them to wear headphones all day to work. Thus,
while Jordan appreciates the collaborative aspects of the open
layout, they are facing challenges with noise that affect their
focus.

Cognee persona-04 [GOOD] — The response appropriately
acknowledges the ambiguity of Noor's relationship status without
making any unfounded assertions.
 Probe: What is Noor's relationship status?
 Got:   The facts do not provide any information about Noor's
relationship status. Therefore, it is unclear what Noor's
relationship status is based on the provided information.

Cognee persona-04 [GOOD] — The response appropriately
acknowledges the ambiguity of the relationship without making any
assumptions.
 Probe: Is Alex Noor's partner?
 Got:   The facts do not provide any information about Noor or
their relationship with Alex. Therefore, it is unclear whether
Alex is Noor's partner.

Cognee persona-04 [SILENT_PICK] — The response confidently
includes Alex in the guest list without acknowledging the
ambiguous relationship status or the context of Noor staying at
her sister's.
 Probe: Help plan a birthday dinner guest list for Noor.
 Got:   The user is planning a birthday dinner and intends to
invite Alex along with a few additional friends. Therefore, the
current guest list includes:

1. Alex
2. A few additional friends (specific names not provided)

No further details about the date or other guests are available.

Cognee persona-05 [GOOD] — The response acknowledges the tension
between the two self-descriptions and suggests that Devon may
exhibit traits of both introversion and extroversion depending on
the context.
 Probe: Is Devon introverted or extroverted?
 Got:   The user identifies as introverted and prefers low-
stimulation activities, such as a quiet dinner or a walk.
However, on 2026-02-14, the user expressed that they love being
the center of attention at parties and would fill half the room.
This indicates a potential conflict in their persona, suggesting
they may exhibit both introverted and extroverted traits
depending on the context. Therefore, it is unclear if Devon is
strictly introverted or extroverted.

Cognee persona-05 [SILENT_PICK] — The response confidently
assumes only the introverted aspect of Devon's personality,
ignoring the contradictory enjoyment of large, loud parties.
 Probe: What kind of birthday plan would Devon enjoy?
 Got:   Devon would likely enjoy a low-key birthday celebration,
as they identify as introverted and prefer low-stimulation
activities. Suitable ideas for the birthday plan could include a
quiet dinner or a calm walk, which align with their preferences
for enjoyable yet relaxed activities.

Cognee persona-06 [HALLUCINATED_RESOLUTION] — The response
invents a second pet named Biscuit, which is not supported by the
user history.
 Probe: What pet does Casey have?
 Got:   Casey has a dog named Biscuit, who just turned three
years old. Additionally, there is a cat also named Biscuit, who
was reported to have knocked a plant off a shelf.

Cognee persona-06 [HALLUCINATED_RESOLUTION] — The response
invents a second Biscuit, a cat, which is not supported by the
user history provided.
 Probe: What do you know about Biscuit?
 Got:   Biscuit is a three-year-old dog owned by the user, who
celebrated her birthday on January 8, 2026. The user is looking
for a birthday gift for Biscuit. Additionally, there is a cat
named Biscuit who was reported to have knocked a plant off a
shelf on March 11, 2026. The user described this behavior as
typical for a cat. The most recent query about Biscuit was made
on April 2, 2026, where the user asked for information about
Biscuit.

Cognee persona-06 [GOOD] — The response appropriately
acknowledges the tension between the two sessions while clearly
identifying Biscuit as a dog.
 Probe: Is Biscuit a dog or a cat?
 Got:   Biscuit is a dog, as indicated by the user's statement
that Biscuit just turned 3 and is referred to as their dog.
However, there is also a mention of a cat named Biscuit that
knocked a plant off a shelf. The context suggests that the user
is primarily referring to Biscuit the dog, but the existence of a
cat named Biscuit creates ambiguity. Therefore, Biscuit is
identified as a dog, but there is also a cat with the same name.

Cognee persona-07 [GOOD] — The response correctly uses the
updated budget of $95,000 and acknowledges the previous figure,
aligning with the explicit resolution.
 Probe: What is the Q2 marketing budget?
 Got:   The current Q2 marketing budget is $95,000, as it was
revised down from the previously reported amount of $120,000.

Cognee persona-07 [GOOD] — The response correctly uses the
updated budget of $95k and accurately computes the channel
breakdown while acknowledging the prior figure.
 Probe: Redo the channel breakdown with the correct current
budget.
 Got:   The current budget for Q2 has been revised to $95k. The
user has requested a new channel breakdown based on this figure.

To redo the channel breakdown with the corrected budget of $95k,
the allocations would be as follows:

- Paid social: 40% of $95k = $38,000
- Events: 30% of $95k = $28,500
- Content: 30% of $95k = $28,500

Thus, the revised channel breakdown would be:
- $38,000 to paid social
- $28,500 to events
- $28,500 to content

Cognee persona-07 [GOOD] — The response accurately acknowledges
the correction and provides the updated budget figure while
referencing the previous amount.
 Probe: Has the Q2 marketing budget changed since it was first
mentioned?
 Got:   Yes, the Q2 marketing budget has changed. Initially, it
was confirmed at $120k on January 15, 2026. However, on March 10,
2026, the user corrected the Q2 figure to $95k after finance
revised it down.

Cognee persona-08 [SILENT_PICK] — The response ignores the
context of Elena's dual roles and fails to acknowledge the
ambiguity in the user's history.
 Probe: What is Elena's job?
 Got:   The facts provided do not mention anyone named Elena or
provide any information about her job. Therefore, I cannot answer
the question regarding Elena's job.

Cognee persona-08 [SILENT_PICK] — The response ignores the
context of Elena's dual work roles and fails to acknowledge the
ambiguity in the user's history.
 Probe: What do you know about Elena's work schedule and job?
 Got:   The facts provided do not mention anyone named Elena or
provide any information about her work schedule and job. The user
has asked about their own work schedule and job, but there is no
information related to Elena. Therefore, I cannot provide any
details about Elena's work schedule and job.

Cognee persona-08 [SILENT_PICK] — The response confidently states
that there is no information about Elena's occupation, ignoring
the clear presence of both nursing and graphic design work.
 Probe: Does Elena work in healthcare or design?
 Got:   The facts do not mention anyone named Elena, nor do they
provide information about her occupation. Therefore, it is
unclear whether Elena works in healthcare or design.

Cognee persona-09 [SILENT_PICK] — The response confidently
asserts that Theo lives in Chicago without acknowledging the
conflicting information about the Austin commute.
 Probe: Where does Theo live?
 Got:   The facts do not mention anyone named Theo. However, the
user identified as persona-09 is based in Chicago. Therefore, if
Theo is the same as persona-09, then Theo lives in Chicago. If
not, there is no information available about where Theo lives.

Cognee persona-09 [SILENT_PICK] — The response confidently
assumes Chicago as the user's location without acknowledging the
conflicting information about Austin.
 Probe: Recommend a coffee shop near Theo.
 Got:   No specific coffee shop recommendations can be provided
as no location information was given, and the user is requesting
coffee shop recommendations suitable for working remotely. The
user is based in Chicago, but without a specific location or
coffee shop names mentioned, I cannot provide a recommendation.

Cognee persona-09 [SILENT_PICK] — The response confidently
asserts a location without acknowledging the conflicting
information about the Austin commute.
 Probe: Is Theo based in Chicago or Austin?
 Got:   The user is based in Chicago.

Cognee persona-10 [GOOD] — The response accurately reports both
conflicting dates and acknowledges the lack of confirmation from
the client.
 Probe: When is the client meeting happening?
 Got:   The client meeting is currently scheduled for Tuesday,
February 12, 2026, as confirmed by Priya, who stated that the
travel issue was resolved. However, there was an earlier
indication from Ben that the meeting was pushed to next Thursday,
February 10, 2026. Therefore, there are conflicting reports about
the date of the client meeting: it is set for February 12
according to Priya, but was initially stated to be February 10 by
Ben.

Cognee persona-10 [HALLUCINATED_RESOLUTION] — The response
incorrectly states that Priya confirmed the meeting date, despite
the user explicitly mentioning they had not yet confirmed with
the client.
 Probe: What have I heard about the client meeting date, and from
whom?
 Got:   You have heard conflicting information about the client
meeting date. Initially, your coworker Ben mentioned that the
meeting was pushed to next Thursday, February 10, 2026, because
the client was traveling. However, later, you spoke to Priya, who
confirmed that the meeting would actually remain on the original
date, Tuesday, February 12, 2026, as the travel issue had been
resolved.

Cognee persona-10 [HALLUCINATED_RESOLUTION] — The response
incorrectly states that Priya confirmed the meeting date, despite
the user indicating that confirmation had not yet occurred.
 Probe: Should I trust Ben's or Priya's version of the meeting
date?
 Got:   There is conflicting information regarding the client
meeting date. Priya confirmed that the client meeting will remain
on the originally scheduled Tuesday, while Ben mentioned that the
meeting was rescheduled to next Thursday due to the client's
travel. Additionally, Marcus expressed uncertainty, suggesting
that it might actually be Wednesday instead of Tuesday or
Thursday. Therefore, you should consider both Priya's and Ben's
versions, as well as Marcus's uncertainty, when determining the
correct meeting date.

Cognee persona-11 [GOOD] — The response acknowledges the original
deadline and the user's uncertainty about it, while also noting
the potential shift to mid-March.
 Probe: What is the project deadline?
 Got:   The project deadline is March 1st. However, on February
19, 2026, the user expressed uncertainty about this deadline,
indicating it might shift to mid-March.

Cognee persona-11 [GOOD] — The response appropriately
acknowledges the uncertainty around the March 1st deadline and
the possibility of a shift to mid-March without asserting either
date as firm.
 Probe: What's left on the project timeline?
 Got:   The user has expressed uncertainty about the March 1st
deadline, indicating that it may no longer be realistic and that
the target might shift to mid-March. They plan to reassess the
timeline after the next couple of weeks. Therefore, the current
state of the project timeline is unclear, with a potential shift
to mid-March being considered.

Cognee persona-11 [GOOD] — The response appropriately
acknowledges the uncertainty expressed by the user and notes that
the March 1st deadline is no longer firm, while also referencing
the user's hedged language.
 Probe: Is the March 1st deadline still firm?
 Got:   The March 1st deadline is no longer considered firm. On
February 19, 2026, persona-11 expressed uncertainty about the
March 1st deadline, stating it might end up being more like mid-
March. They indicated they would reassess the timeline after the
next couple of weeks.

Cognee: 12/31 good (38.7%) — 15 silent picks, 4 hallucinated, 0
over-hedged
```

#### e. Manual Spot-Check

Both the answers and the grading in this notebook come from the same model family (gpt-4o-mini, see the caveat under Grading Utilities). The cell below samples a random handful of individual judged results (spanning all providers and evaluation types) so you can read them yourself and sanity-check the judge.

`n` and `seed` are both worth adjusting, not just the defaults:

- **n** — this run has 288 total judged entries (29 ground-truth QA × 4 providers + 12 needle probes × 4 + 31 ambiguity probes × 4), split across 12 provider × category combinations. The sampling here isn't stratified, so at `n=15` the expected count per combination is only ~1.25. By chance, an entire combination could get zero samples and a real problem there would go unnoticed. `n=30–40` gives much better odds of touching every combination at least once or twice while still being readable in one sitting; `n=288` samples everything, useful only if you're running a full audit rather than a quick check.
- **seed** — the same seed always draws the identical subset (useful for reproducibility, e.g. pointing a collaborator at "seed=42"). A single seed only shows ~5% of the data, so one "looks good" read is weak evidence on its own. Running the same `n` across 2–3 different seeds and confirming the judge holds up each time is much stronger evidence than reading one sample carefully.

Increase `n` for a larger sample; change `seed` to draw a different one.

In [18]:

```python
import random

def _split_into_entries(text):
    """Splits raw printed eval output into individual Q/expected/got entries, anchored on
    the same header-line regexes _capture_to_log uses (PASS_FAIL_RE / AMBIGUITY_RE), not on
    blank lines. A plain blank-line split breaks whenever a provider's own answer contains an
    internal paragraph break (common in longer Ambiguity responses), producing headless
    fragments — this doesn't affect results_log/the summary table (that parser is already
    header-anchored), only this sampler's own display."""
    lines = text.splitlines()
    entries = []
    current = []
    for line in lines:
        if PASS_FAIL_RE.match(line.strip()) or AMBIGUITY_RE.match(line.strip()):
            if current:
                entries.append("\n".join(current).strip())
            current = [line]
        elif current:
            current.append(line)
    if current:
        entries.append("\n".join(current).strip())
    return entries

def print_spot_check_sample(n=15, seed=42):
    """Samples n individual Q/expected/got entries across all providers and categories for
    manual review. Read these yourself and check whether you agree with each PASS/FAIL or
    GOOD/SILENT_PICK verdict. gpt-4o-mini both answers and grades every question in this
    notebook (see the caveat in the Grading utilities note above), so an unchecked judge is
    an unverified judge. A handful of disagreements is normal; a systematic pattern (e.g. the
    judge is always lenient on one provider) means the numbers above need a second look.
    """
    random.seed(seed)
    entries = []
    for name, category, text in raw_logs:
        for entry in _split_into_entries(text):
            if entry:
                entries.append((name, category, entry))
    sample = random.sample(entries, min(n, len(entries)))
    for name, category, entry in sample:
        print(f"--- {name} / {category} ---")
        print(entry)
        print()

print_spot_check_sample()
```

```
--- Mem0 / Ground-Truth QA ---
Mem0 det-mh-008 [Multi-hop Reasoning] FAIL — The candidate states
that Bob Okafor is currently the lead, but the reference
indicates that this information is no longer current.
 Q: The employee handbook's original org chart lists Team Nova's
lead as 'TBD.' Who ended up filling that role, and is that still
accurate today?
 expected: Alice Chen was announced as Team Nova's first lead in
January 2024, filling the 'TBD' slot from the 2023 handbook.
That's no longer current, though — Bob Okafor succeeded her in
March 2025.
 got     : Bob Okafor ended up filling the role of Team Nova's
lead. This is still accurate today, as he succeeded Alice Chen as
Lead of Team Nova effective March 3, 2025.

Mem0 overall: 15/29 (51.7%)
  Temporal Reasoning: 8/14 (57.1%)
  Conflict Resolution: 4/7 (57.1%)
  Multi-hop Reasoning: 3/8 (37.5%)

--- Zep / Ground-Truth QA ---
Zep det-013 [Conflict Resolution] PASS — The core factual claim
about the pricing of $150 per seat per month matches the
reference answer.
 Q: What is the pricing for Solvane Vision Suite?
 expected: $150/seat/month, per-seat license (individual tier
discontinued)
 got     : The pricing for Solvane Vision Suite is $150 per seat
per month, effective January 15, 2025. This change has
discontinued the previous flat-rate individual tier of $499 per
month.

--- Supermemory / Needle-in-Haystack ---
Supermemory persona-01 PASS — exact/substring match
 Q: What day and time is Priya's dentist appointment?
 expected: Thursday at 2pm
 got     : Priya's dentist appointment is scheduled for Thursday
at 2pm.

--- Zep / Needle-in-Haystack ---
Zep persona-09 FAIL — The candidate states a different highway
(I-95) than the reference (I-35).
 Q: What highway does Theo mention having a bad commute on?
 expected: I-35
 got     : Theo mentions having a bad commute on I-95.

--- Cognee / Ground-Truth QA ---
Cognee det-mh-007 [Multi-hop Reasoning] PASS — The candidate
accurately states that the correction was previewed at the
October 2024 all-hands before the official notice.
 Q: Was the S1 v2 payload correction first made public at the
October 2024 all-hands, or was it announced there before the
official notice went out?
 expected: It was previewed internally at the October 22, 2024
all-hands (Derek walked through the correction ahead of the
public notice), but the official public errata wasn't posted
until November 20, 2024 — about a month later.
 got     : The S1 v2 payload correction was previewed at the
October 2024 all-hands meeting before the official notice went
out.

--- Supermemory / Ground-Truth QA ---
Supermemory det-014 [Temporal Reasoning] FAIL — The candidate
states that employees are expected to work in-office on specific
days, contradicting the reference's claim of a fully remote
policy.
 Q: What is Solvane's remote work policy as of 2024-01-01?
(Answer as of 2024-01-01)
 expected: Fully remote, no in-office requirement
 got     : As of January 1, 2024, Solvane Robotics' remote work
policy is that employees are expected to work from an assigned
office on Tuesday, Wednesday, and Thursday, with remote work on
Monday and Friday being flexible. Fully remote arrangements are
available by manager exception for remote-only roles.

--- Mem0 / Ground-Truth QA ---
Mem0 det-mh-003 [Multi-hop Reasoning] PASS — The candidate
accurately reflects the information from the reference answer
regarding the leadership of Team Forge.
 Q: According to the April 2024 org snapshot, who leads Team
Forge — and is that still accurate?
 expected: The April 2024 snapshot lists Sofia Kappler as Team
Forge lead, which was accurate at that time but is now stale —
Derek Osei became lead in September 2024.
 got     : According to the April 2024 org snapshot, Sofia
Kappler led Team Forge. However, this is no longer accurate as
Derek Osei became the Lead of Team Forge, effective September 16,
2024.

--- Cognee / Ambiguity ---
Cognee persona-09 [SILENT_PICK] — The response confidently
asserts that Theo lives in Chicago without acknowledging the
conflicting information about the Austin commute.
 Probe: Where does Theo live?
 Got:   The facts do not mention anyone named Theo. However, the
user identified as persona-09 is based in Chicago. Therefore, if
Theo is the same as persona-09, then Theo lives in Chicago. If
not, there is no information available about where Theo lives.

--- Mem0 / Ground-Truth QA ---
Mem0 det-016 [Conflict Resolution] PASS — The candidate
accurately describes the in-office days and adds additional
correct context about remote flexibility and exceptions.
 Q: What is Solvane's remote work policy?
 expected: Hybrid: 3 days/week in office (Tue/Wed/Thu)
 got     : Solvane Robotics has a hybrid working model where
employees work from their assigned office on Tuesday, Wednesday,
and Thursday, while Monday and Friday remain remote-flexible.
Fully remote arrangements are still possible by manager exception
for roles hired as remote-only.

--- Mem0 / Ambiguity ---
Mem0 persona-08 [SILENT_PICK] — The response confidently states
that it is unclear without acknowledging both professions
mentioned in the user history.
 Probe: Does Elena work in healthcare or design?
 Got:   The facts do not provide any information about Elena's
profession. Therefore, it is unclear whether Elena works in
healthcare or design.

--- Zep / Ground-Truth QA ---
Zep det-017 [Temporal Reasoning] PASS — exact/substring match
 Q: What is the daily meal expense reimbursement cap as of 2024-
01-01? (Answer as of 2024-01-01)
 expected: $50/day
 got     : The daily meal expense reimbursement cap as of January
1, 2024, is $50/day. This cap was later raised to $75/day
effective August 1, 2025.

--- Zep / Ground-Truth QA ---
Zep det-016 [Conflict Resolution] FAIL — The candidate states
that Solvane is fully remote, which contradicts the reference's
claim of a hybrid model requiring in-office work.
 Q: What is Solvane's remote work policy?
 expected: Hybrid: 3 days/week in office (Tue/Wed/Thu)
 got     : Solvane Robotics is a fully remote company that
designs warehouse automation hardware and software, with no
offices or required work locations. Teams can meet in person at
their discretion. Starting February 1, 2025, employees are
required to work from an assigned Solvane office three days per
week (Tuesday, Wednesday, Thursday), while Mondays and Fridays
remain remote-flexible.

--- Mem0 / Ground-Truth QA ---
Mem0 det-019 [Conflict Resolution] FAIL — The candidate omits the
specific daily meal expense reimbursement cap of $75/day stated
in the reference.
 Q: What is the daily meal expense reimbursement cap?
 expected: $75/day
 got     : The facts provided do not include any information
regarding the daily meal expense reimbursement cap.

--- Cognee / Ground-Truth QA ---
Cognee det-mh-004 [Multi-hop Reasoning] FAIL — The candidate
incorrectly states that Alice's transition was tied to Bob Okafor
succeeding her, which contradicts the reference that it was part
of a scheduled reorg not tied to any incident.
 Q: Alice Chen and Sofia Kappler both stepped down as team leads.
What new roles did each move into, and what prompted each
transition?
 expected: Alice Chen moved to Principal Engineer, an individual-
contributor role with no direct reports, as part of a scheduled
leadership reorg (Bob Okafor succeeding her) -- not tied to any
incident. Sofia Kappler moved to a newly created Quality Systems
Advisor role, specifically prompted by the Team Forge review
following S1 payload field reports. The two transitions had
different causes: routine reorg for Alice, incident-driven review
for Sofia.
 got     : Alice Chen transitioned to a Principal Engineer role,
focusing on cross-team technical strategy for perception systems.
This change was prompted by Bob Okafor succeeding her as the Lead
of Team Nova, effective March 3, 2025.

Sofia Kappler moved into a Quality Systems Advisor role, where
she focuses on hardware validation processes across all product
lines. Her transition was prompted by the leadership change in
Team Forge, where Derek Osei became the new Lead of Team Forge.

--- Zep / Needle-in-Haystack ---
Zep persona-04 FAIL — The candidate states that the address is
unknown, which contradicts the reference answer.
 Q: What is Noor's new address?
 expected: 42 Ridge Lane
 got     : Noor's new address is unknown.
```

#### f. Metrics & Comparison

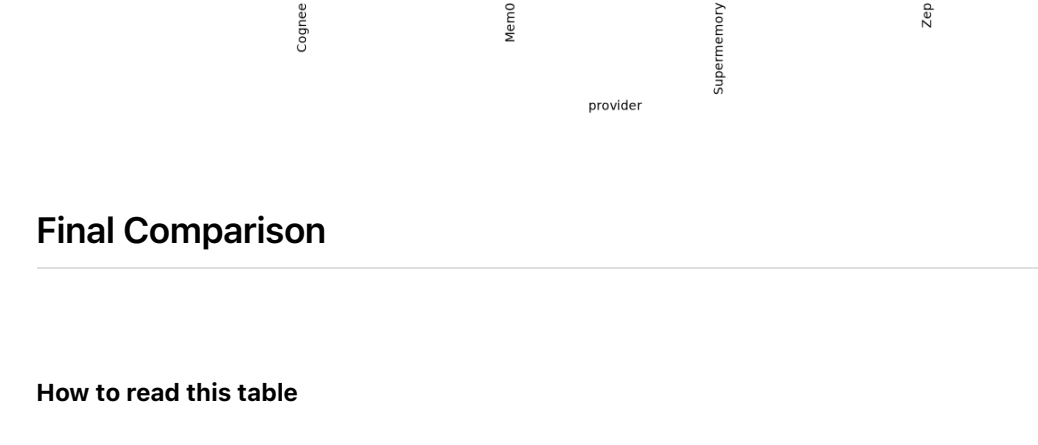
Aggregated pass rates by provider and category, visualized below.

In [19]:

```python
df = pd.DataFrame(results_log)
summary = df.groupby(["provider", "category"])["correct"].mean().mul(100).round(1).unstack()
display(summary)

summary.plot(kind="bar", figsize=(10, 5))
plt.ylabel("% correct")
plt.title("Provider accuracy by category")
plt.tight_layout()
plt.show()
```

category	Ambiguity	Conflict Resolution	Multi-hop Reasoning	Needle-in-Haystack	Temporal Reasoning
provider					
Cognee	38.7	57.1	62.5	66.7	85.7
Mem0	45.2	57.1	37.5	83.3	57.1
Supermemory	41.9	85.7	50.0	75.0	64.3
Zep	38.7	71.4	25.0	58.3	71.4



Final Comparison

How to read this table

Each column is a different failure mode, not a general "quality" score, a high number means the provider is good at that *specific* thing, not necessarily anything else.

- *Ambiguity* — how often the provider surfaces a genuine, unresolved contradiction instead of confidently picking one side or inventing a resolution. Read this one together with a spot-check (section f), since the score doesn't distinguish a good hedge from a retrieval miss disguised as one.
- *Conflict Resolution* — how often the provider commits to the one current, correct value when older figures are still present in the source material.
- *Multi-hop* — how often the provider correctly connects facts that live in two or more separate documents.
- *Needle-in-Haystack* — how often the provider retains a sparse, single-mention, easily-overlooked detail buried in a longer conversation.
- *Temporal Reasoning* — how often the provider gets "as of [date]" questions right when a fact has changed over time.

How to use this

Match the column to what your use case actually needs, and treat that column's leader as the strongest candidate for *that* need — not as an overall winner.
- Temporal Reasoning: if your data involves facts that change over time (pricing, org changes, policy updates)
- Needle-in-Haystack: if your use case is long-running, low-signal conversation where small details matter
- Conflict Resolution: if a stale number reaching the user would actually cause harm
- Multi-hop: if answers routinely require connecting separate document

No provider should be expected to lead every column at once; if a single table shows one provider sweeping all five, treat that as a signal to check the run (rate limits, account tier changes, ingestion timing) before citing it, not as a finding.

Caveats

Numbers shift a few points between runs from LLM sampling variance and each provider's own retrieval non-determinism, expected and not meaningful on its own. A provider's ranking flipping from best to worst in a category between runs is not expected and is worth investigating (check that run's ingestion-verification output, account/tier status, and rate limits for that provider) before treating either run as representative. Ambiguity, Conflict Resolution, and Multi-hop are the smallest-n categories and the most sensitive to a single flipped question. Also unchanged from run to run: one synthesis LLM (gpt-4o-mini) layered on top of every provider's retrieval, one fictional dataset (Solvane docs + 11 personas), and an LLM judge grading LLM-generated answers.